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Hard Matter: Fundamentals and Applications

**Structural, Calorimetric, Diffractometric, Spectroscopic and Electrochemical
Characterization of AgI and Gd-Doped CeO₂ systems**

Laboratory Report

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1. Introduction

Silver iodide (AgI) is a useful material for understanding the physical properties of a solid, as it can suffer distinctive changes in its crystal structure in relatively mild experimental conditions, hence giving us insights into the nature of its phase transitions. At low temperatures, AgI is mainly found as β -AgI, having a hexagonal structure usually described as wurtzite-type (Figure 1a). In this arrangement, Ag^+ and I^- ions are organized in a tetrahedral network, meaning that each ion is surrounded by four ions of opposite charge. This structure is considerably ordered and relatively stable, making the mobility of silver ions limited in comparison with higher temperature structures. When β -AgI is heated close to 420 K, it transforms into α -AgI. This phase is more disordered in the silver sublattice, meaning that the I^- ions keep a more defined framework, while Ag^+ ions can move between several available positions. Because of this, α -AgI is considered a classic superionic conductor: it is still a solid, but part of its ionic structure behaves almost like it was liquid¹. For this reason, the $\beta \rightarrow \alpha$ transition is not only a structural change, but also a strong change in ionic transport².

In this work, AgI was studied using different calorimetric, diffractometric and spectroscopic techniques to describe its behavior from multiple points of view. Differential scanning calorimetry (DSC) was used to detect the thermal phase transition; x-ray powder diffraction (XRPD), to confirm the structural change between β -AgI and α -AgI; ultraviolet-visible diffuse reflectance spectroscopy (UV-Vis-DRS), to estimate the band gap; and electrochemical impedance spectroscopy (EIS), to follow how the electrical response and conductivity changes with temperature. Together, these techniques help to connect the phase transition with the optical and transport properties of AgI.

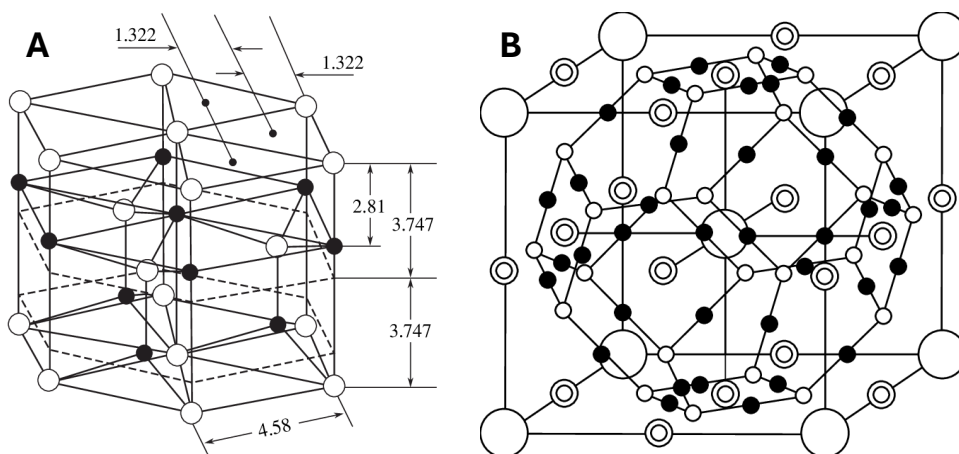


Figure 1: Crystal structures of A) β -AgI³ B) α -AgI⁴. Large empty circles represent I^- ions, while small circles, the different positions and vacancies that can be adopted by Ag^+ ions.

2. Differential Scanning Calorimetry (DSC)

2.1 Principle of Technique

DSC is a thermal analysis technique used to study solid dispersions, such as AgI, by measuring its difference in heat flow (HF) compared to a reference material under the same controlled temperature program, thus allowing to assess the phase transitions as a function of temperature

or time. The calculated HF is based on the instrument characteristics, as well as on internal and/or external standard calibrants. Whenever the sample undergoes a phase transition, heat is absorbed or released, giving an endothermic (positive) or exothermic (negative) peak in the thermogram, respectively. These changes in enthalpy are widely used to characterize the properties of solid-state materials in a rapid manner due to its relatively simple and straightforward sample preparation procedure^{5,6}.

For this purpose, a heating ramp followed by a cooling ramp, was applied at seven different temperature scan rates: 1, 2, 5, 10, 20, 30, and 50 K min⁻¹ (Figure 2, Tables 1-2). This was made to extrapolate the transition parameters to the rate-independent or quasi-equilibrium limit, and to assess kinetic effects on the apparent transition temperature. The signal can be plotted either as a function of time or temperature. However, the time domain is preferred for quantitative analysis. From the integrated peak area, the following quantities were extracted:

- Enthalpy change of the transition ($\Delta_{\text{trans}}H$), per unit mass or per mole.
- Entropy change (ΔS), obtained from the equilibrium condition at the transition temperature: $\Delta_{\text{trans}}S = \Delta_{\text{trans}}H / T_{\text{onset}}$.

The onset temperature, T_{onset} , is defined as the intersection of the pre-transition baseline with the tangent line drawn at the point of maximum slope on the leading edge of the peak. This becomes the best estimate of the true thermodynamic transition temperature, since it is less sensitive to peak broadening than the peak maximum, mainly due to artifacts dependent on scanning rate. On the other hand, a complementary set of measurements was carried out on four independent sample groups, with masses ranging 15.6-34.5 mg, at a fixed heating rate of 10 K min⁻¹ (Table 3), to test the dependence or not on sample size of the recorded signal. Moreover, the readings of the instrument employed for the measurements were verified with a known Indium (In) standard, which has ample evidence for the thermodynamic parameters associated with the phase transition of its melting point. Thus, it was used to correct the enthalpy values obtained during the calorimetric measurements with a scaling factor of 0.8983 based on the relationship between the measured $\Delta_{\text{melting}}H_{\text{In}} = 31.898$ J/g during the heating stage, with respect to the reference value of 28.53 J/g⁷. In a similar way, for the cooling stage was used a correction factor of 0.8883 as the measured $\Delta_{\text{melting}}H_{\text{In}} = -32.116$ J/g.

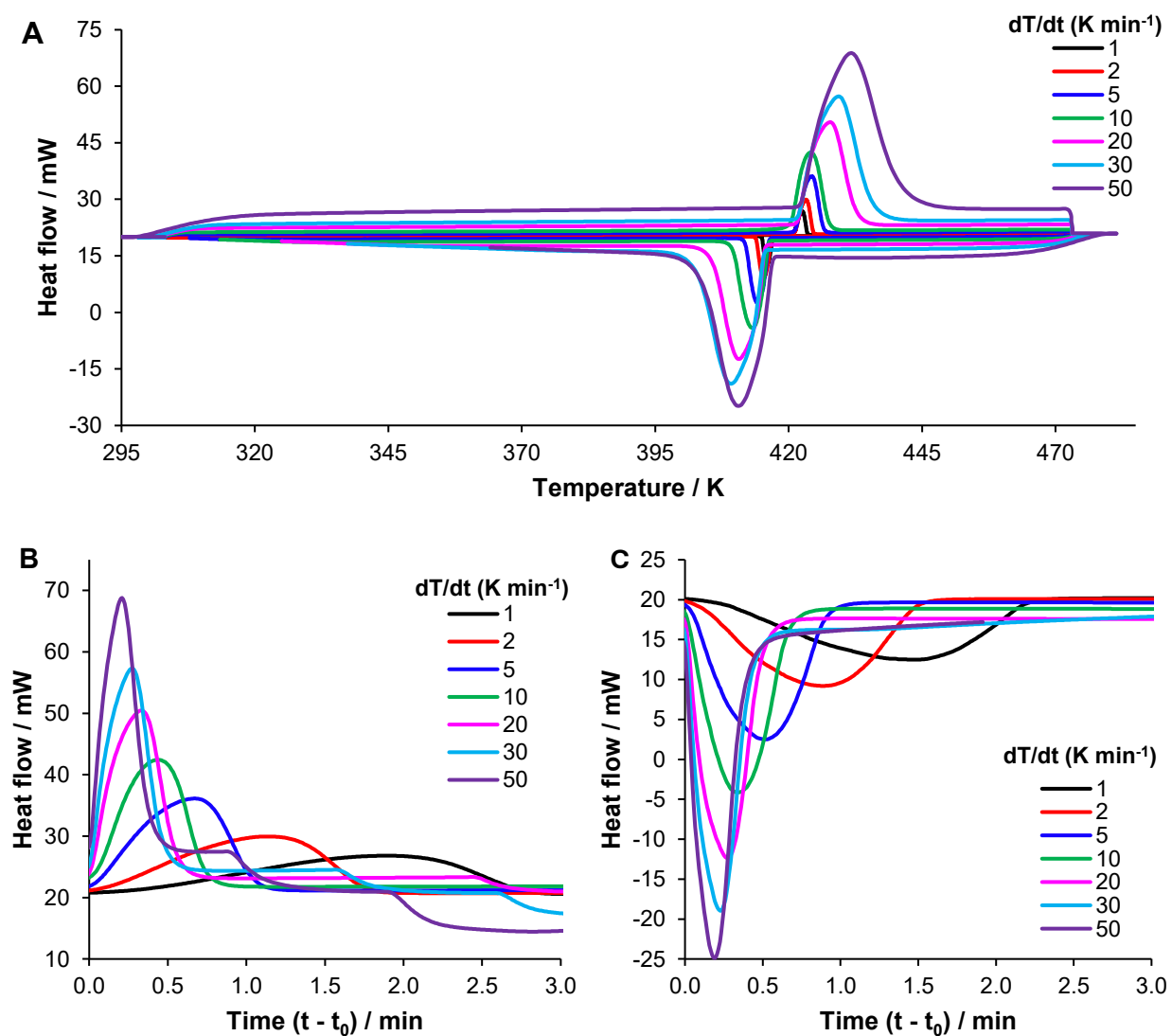


Figure 2. A) DSC thermogram of AgI at different dT/dt rates (1, 2, 5, 10, 20, 30, 50 K min⁻¹). B) Peak shifting and sharpening at increasing dT/dt rates during the heating stage of AgI as a function of time, normalized with respect to the onset time (t_0). C) Peak shifting and sharpening at increasing dT/dt rates during the cooling stage of AgI as a function of time normalized with respect to the onset time (t_0).

2.2 Effect of heating rate on the $\beta \rightarrow \alpha$ transition for AgI.

Table 1. Effect of heating rate on thermodynamic parameters for the $\beta \rightarrow \alpha$ phase transition of AgI.

Scan rate (K min ⁻¹)	T _{onset} (K)	t _{onset} (min)	AUC (mJ)	$\Delta_{tr}H$ corrected (kJ mol ⁻¹)	$\Delta_{tr}S$ (kJ mol ⁻¹ K ⁻¹)
1	420.700	118.850	596.691	5.747	0.01366
2	421.060	60.242	594.754	5.729	0.01361
5	421.210	24.892	596.235	5.743	0.01363
10	420.140	12.975	591.841	5.701	0.01357
20	421.540	7.200	591.820	5.701	0.01352
30	421.750	5.233	591.132	5.694	0.01350
50	422.250	3.658	584.984	5.635	0.01334
Mean:	421.2 ± 0.7	-	592 ± 4	5.71 ± 0.04	0.0135 ± 0.0001

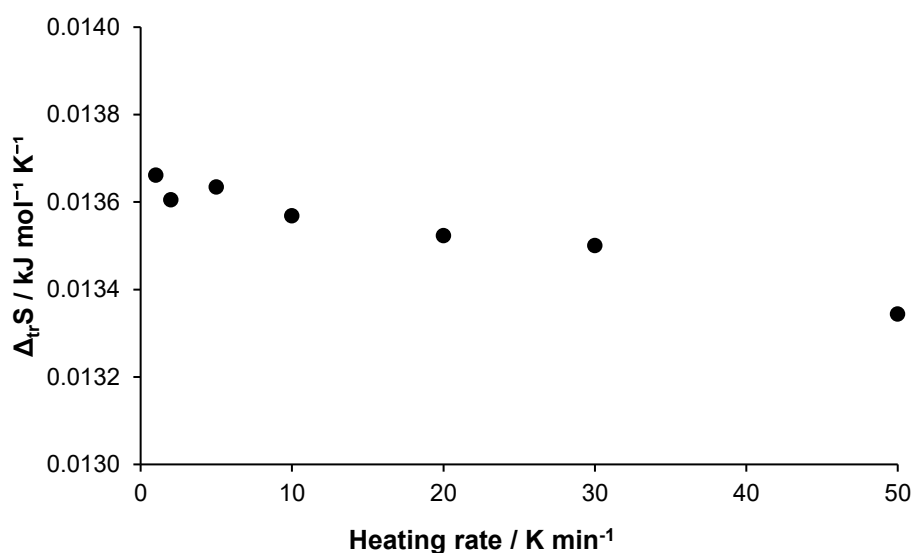


Figure 3. Transition entropy ($\Delta_{tr}S$) as a function of heating rate

The $\Delta_{tr}S$ decreases steadily following a linear trend, thus revealing an inversely proportional to the heating rate. Because $\Delta_{tr}S = \Delta_{tr}H/T_{onset}$, this downward trend reflects the combined effect of a slightly decreasing $\Delta_{tr}H$ and a slightly increasing T_{onset} with rate (Figures 3-5): both changes push $\Delta_{tr}S$ in the same (decreasing) direction as scan rate rises.

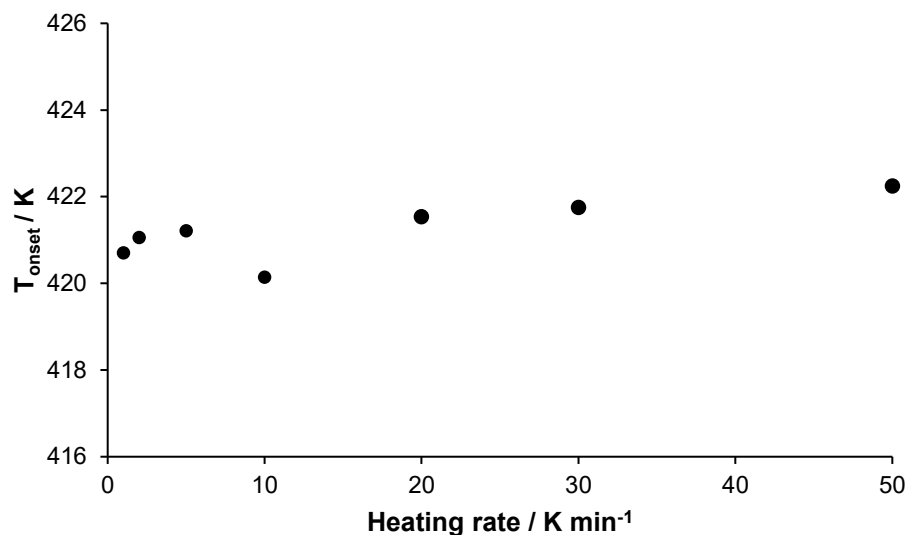


Figure 4. Onset temperature (T_{onset} , K) as a function of heating rate

The T_{onset} increases with heating rate, from about 420.7 K at 1 K min^{-1} to about 422.2 K at 50 K min^{-1} . This could be associated with the thermal lag of the DSC readings: at faster rates the sample's true temperature increasingly gets delayed behind the programmed furnace temperature, so the transition appears to occur later, even though the intrinsic equilibrium transition temperature is unchanged. In addition, figure 4 shows how this effect is more evident when the change happens at low heating rates, but after it goes beyond 20. K min^{-1} , the delay increases following a linear trend.

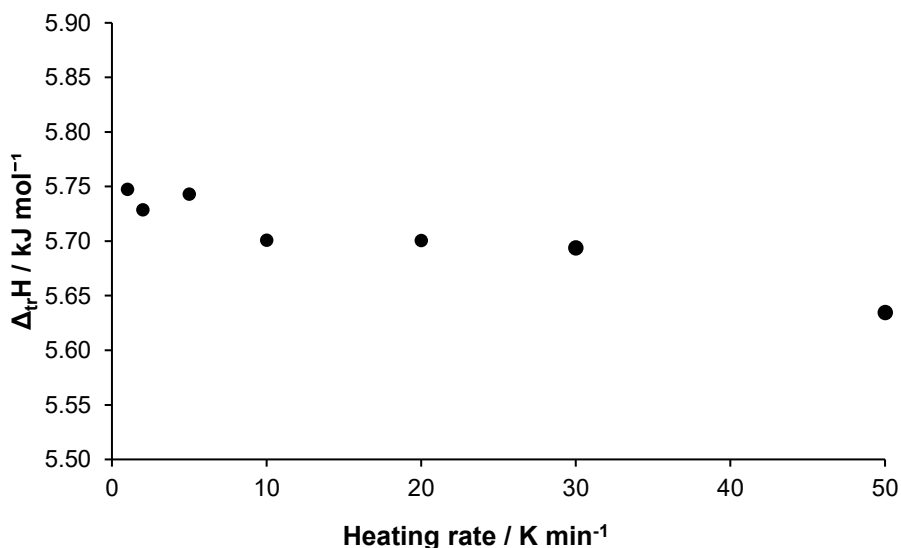


Figure 5. Transition enthalpy ($\Delta_{\text{tr}}H$) vs. heating rate

The $\Delta_{\text{tr}}H$ decreases slightly but consistently with increasing heating rate ($\approx 2\%$ from the slowest to the fastest rate). This small negative dependence is again consistent with thermal lag and with peak broadening at high scan rates, which can lead to slightly less complete integration of the true peak area.

2.3 Effect of cooling rate on the $\alpha \rightarrow \beta$ transition for AgI.

Table 2. Effect of cooling rate on thermodynamic parameters for the $\alpha \rightarrow \beta$ phase transition of AgI.

Scan rate (K min ⁻¹)	T _{onset} (K)	t _{onset} (min)	AUC (mJ)	$\Delta_{tr}H$ corrected (kJ mol ⁻¹)	$\Delta_{tr}S$ (kJ mol ⁻¹ K ⁻¹)
1	417.218	228.211	-600.580	-5.746	-0.01377
2	416.896	115.410	-603.075	-5.770	-0.01384
5	416.462	47.622	-603.710	-5.776	-0.01387
10	416.283	24.971	-605.933	-5.797	-0.01393
20	415.763	13.656	-607.938	-5.816	-0.01399
30	415.684	9.874	-609.891	-5.835	-0.01404
50	417.062	6.884	-609.280	-5.829	-0.01398
Mean:	416.5 ± 0.6	-	-606 ± 3	-5.80 ± 0.03	-0.01392 ± 0.00009

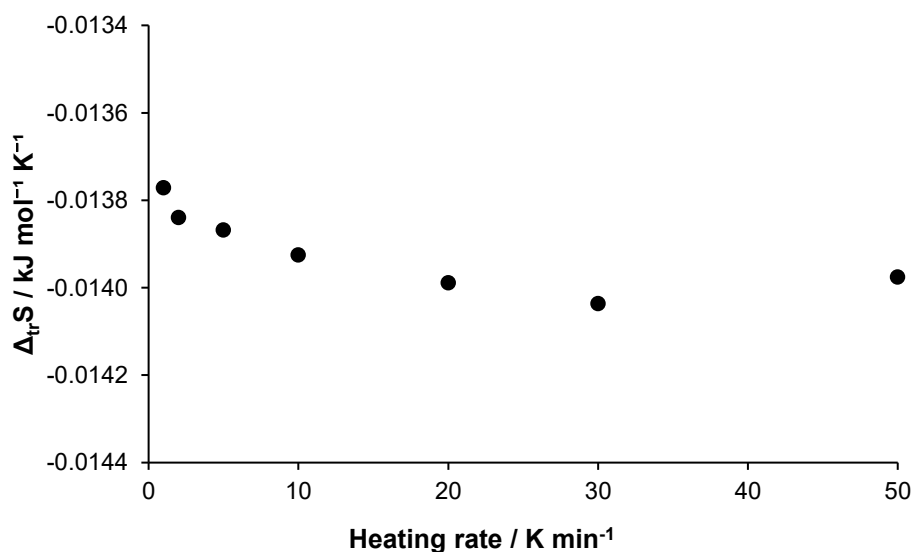


Figure 6. Transition entropy ($\Delta_{tr}S$, negative/exothermic) as a function of the cooling rate.

The $\Delta_{tr}S$ becomes progressively more negative (Table 2, Figure 6) as the cooling rate increases, i.e. $|\Delta_{tr}S|$ increases with rate. The mirror image of the heating behavior, again consistent with kinetic effects driving the transition further from equilibrium at faster rates.

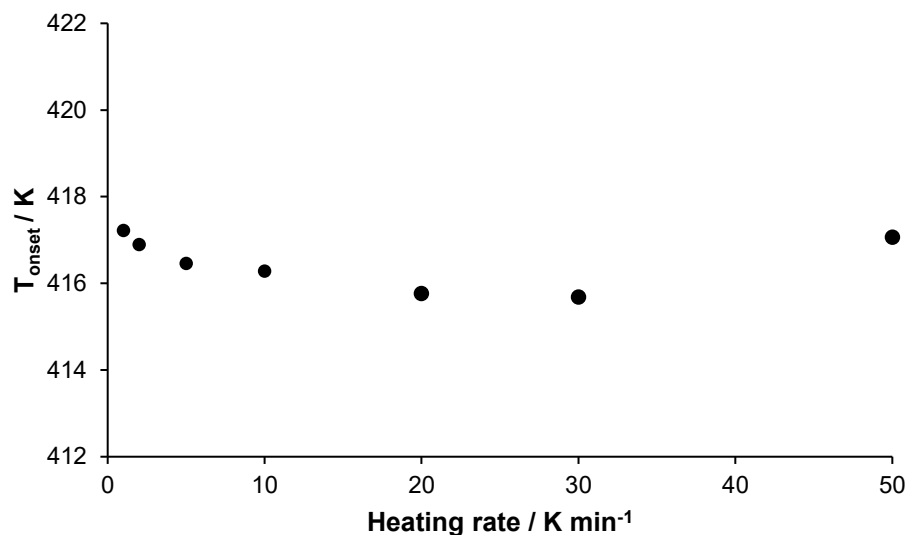


Figure 7. Onset temperature (T_{onset} , K) as a function of cooling rate.

In clear contrast to the heating series, T_{onset} on cooling decreases with increasing rate, dropping from about 417.2-416.9 K at the slowest rates to about 415.7-417.1 K at the faster rates (Figure 7). This is the behavior expected for a first-order transition with thermal hysteresis: faster cooling leaves less time for nucleation of the new (β) phase, so the transition becomes increasingly supercooled. Combined with the heating data, the heating-cooling hysteresis gap widens with scan rate, from roughly 5 K at the slowest rates to roughly 11-12 K at the fastest rates.

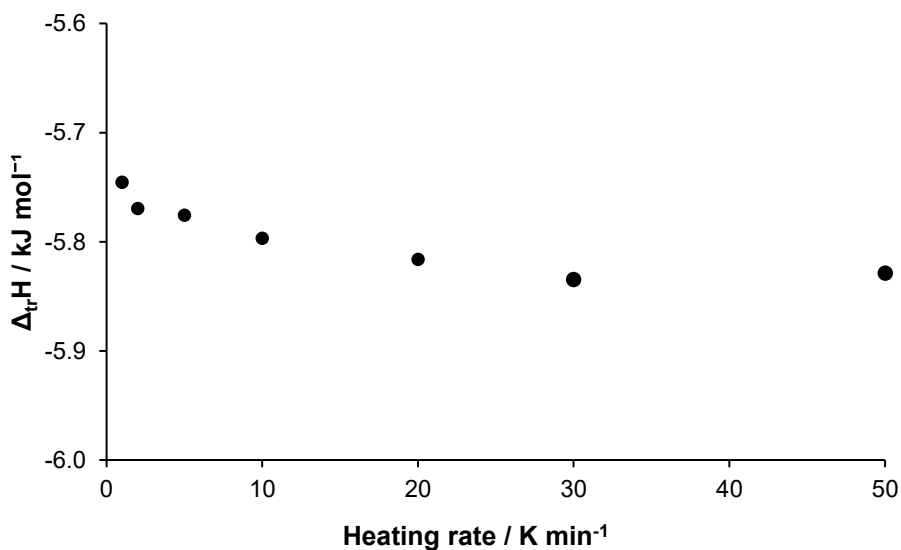


Figure 8. Transition enthalpy ($\Delta_{\text{tr}}H$, negative/exothermic) as a function of the cooling rate.

When both heating and cooling rate curves are compared, it can be observed that the T_{onset} , $\Delta_{\text{tr}}H$ and $\Delta_{\text{tr}}S$ display mirror-like behaviors depending on which step of the cycle is the sample: with $\Delta_{\text{tr}}H$ and $\Delta_{\text{tr}}S$ decreasing and T_{onset} increasing progressively during the heating curve, while the opposite happens when the system is returning to the room temperature during the cooling phase (Figure 8). This could be explained by kinetic lag/supercooling moving the apparent transition

temperature away from the true equilibrium value as the experiment is driven faster in either direction.

Both $\Delta_{tr}H$ and $\Delta_{tr}S$ shift only slightly, yet the absolute value of the heating and cooling enthalpies (Tables 1-2) and entropies stay consistent around their respective average values (5.7 - 5.8 kJ mol⁻¹ and 0.0135-0.0139 kJ mol⁻¹ K⁻¹), having opposite sign and confirming the transition is thermodynamically reversible, which is also in accordance with the values reported in the literature.

The hysteresis gap between heating and cooling onset temperatures widens with scan rate, a hallmark of a first-order transition, whose forward and reverse kinetics are rate-limited by nucleation, which could mainly be promoted by defects, migration of ionic species, etc.

2.4 Effect of Sample Mass

Table 3. Sample mass and fixed heating rate (10 K min⁻¹) of AgI for DSC groups.

Group	Mass (mg)	Heating Rate (K min ⁻¹)	T _{onset} (K)	t _{onset} (min)	AUC (mJ)	$\Delta_{tr}H$ corrected (kJ mol ⁻¹)	$\Delta_{tr}S$ (kJ mol ⁻¹ K ⁻¹)
1	34.5	10	420.36	13.000	879.100	5.351	0.01273
2	24.7	10	420.29	12.992	644.580	5.480	0.01304
3	26.5	10	421.85	13.142	724.757	5.743	0.01361
4	15.6	10	420.59	13.025	406.249	5.468	0.01300
Mean:	25 ± 8	10	420.8 ± 0.7	13.04 ± 0.07	664 ± 197	5.5 ± 0.2	0.0131 ± 0.0004

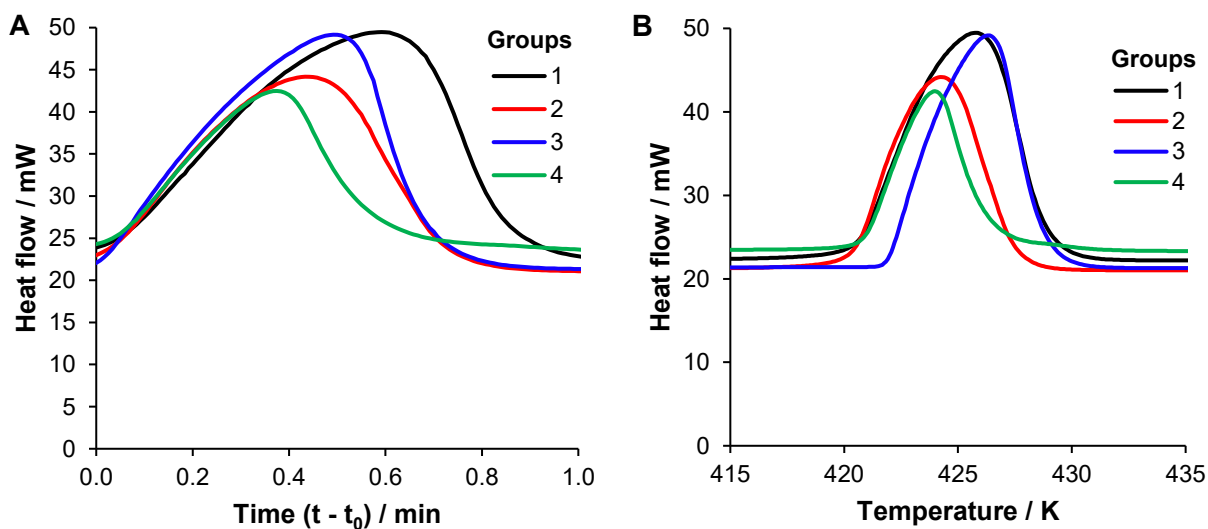


Figure 9. Heat flow (mW) of variable amounts of AgI vs. time from transition onset ($t-t_0$) (A) and vs. temperature (B) for Groups 1-4.

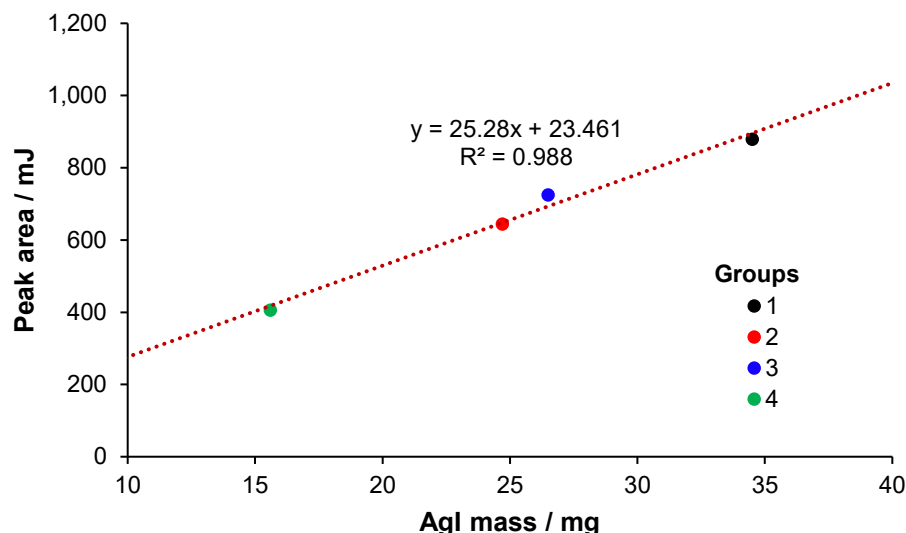


Figure 10. Peak area (mJ) vs. mass of AgI (mg) for Groups 1-4.

Both representations in figures 9A and 9B show the same essential shape for all four groups: a flat baseline (≈ 20 mW) followed by a positive peak discontinuity in the heat-flow level as the sample passes through the $\beta \rightarrow \alpha$ transition, confirming the phase transition is detected consistently regardless of sample size.

The onset of the rise occurs in a narrow window (420.8 ± 0.7 K, consistent with the heating T_{onset} values in section 2.2), and group 3 (blue) shows a slightly later onset than the other three groups which could be attributed to the random error associated with the analyst. Nonetheless, the onset temperatures and times, as well as the transition enthalpy and entropy are all within narrow consistent windows, with only the area of the peak (Table 3). Comparing this with the masses of the samples shows that, as expected, these thermodynamic properties do not scale or correlate with the quantity of samples as enthalpy and entropy are state functions, and as such their values per unit of mol or mass should remain constant, and depend only on the difference between the initial and final state.

This leaves only the area of the peak to show a linear correlation with the amount of sample (Figure 10), which was totally expected, as the resulting enthalpy values obtained from the instrument are expressed in terms of energy units exchanged with the surrounding environment per units of mass (e.g. J/g), based on the equation $A = m K(T) \Delta H_s$, where A is the area of the peak, m the mass of the sample, $K(T)$ is a calibration constant, ΔH_s is the enthalpy of the process per unit of mass.

In summary, the DSC results give a coherent picture of a reversible, first-order $\beta \leftrightarrow \alpha$ phase transition in AgI, located near 421.2 ± 0.7 K (≈ 148 °C) on heating and 416.5 ± 0.6 K (≈ 143 °C) on cooling, with $|\Delta_{\text{tr}}H| \approx 5.7\text{--}5.8$ kJ mol $^{-1}$ and $|\Delta_{\text{tr}}S| \approx 0.0135\text{--}0.0139$ kJ mol $^{-1}$ K $^{-1}$. Extrapolating the heating-onset trend to zero rate gives the most reliable equilibrium transition temperature estimate, ≈ 421.0 K (≈ 147.8 °C), in good agreement with literature values for this superionic transition ($\approx 146\text{--}147$ °C).

3. X-Ray Powder Diffraction (XRPD)

3.1 Principle of the Technique

X-ray Powder Diffraction (XRPD) probes the long-range periodic arrangement of atoms in a crystalline solid. When a monochromatic X-ray beam strikes a powdered crystalline sample, constructive interference occurs at specific angles (2θ) according to Bragg's law, producing a diffraction pattern whose peak positions are determined by the unit-cell dimensions and whose peak widths and intensities are determined by crystallite size, microstrain and structural disorder⁸. For the AgI system, XRPD was used to monitor the structural changes as a function of temperature, to follow the shift of diffraction peaks across the $\alpha \rightarrow \beta$ phase transition, and to extract the lattice parameters and unit-cell volume for both phases.

3.2 Cell parameters of β -AgI

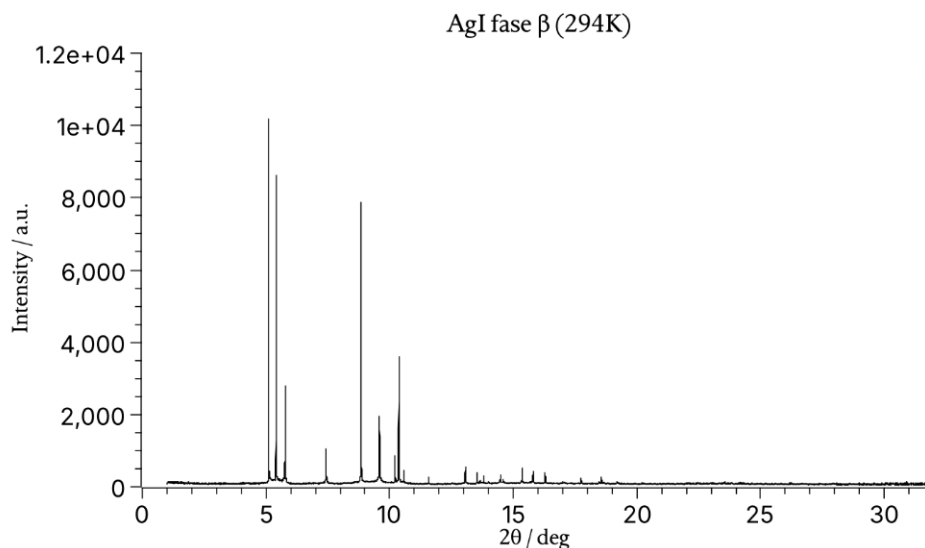


Figure 11. XRPD pattern of β -AgI at 294 K. The diffraction peaks correspond to the low-temperature hexagonal phase.

At low temperatures, from 294 K to 377 K (Table 4), the diffraction peaks correspond to the β phase of AgI (Figure 11). This phase has a hexagonal structure. The position of the peaks changes only slightly with temperature, which means that the crystal structure remains almost the same in this range. Small shifts of the peaks can be related to small changes in the unit cell parameters caused by temperature.

Table 4. XRPD diffraction peaks for β phase of AgI.

h k l	294 K 2 θ / deg	315 K 2 θ / deg	335 K 2 θ / deg	357 K 2 θ / deg	377 K 2 θ / deg
1 0 0	5.10511 $\pm$ 0.00002	5.10492 $\pm$ 0.00002	5.10481 $\pm$ 0.00002	5.10469 $\pm$ 0.00002	5.10485 $\pm$ 0.00002
0 0 2	5.40920 $\pm$ 0.00003	5.40944 $\pm$ 0.00003	5.40985 $\pm$ 0.00003	5.41030 $\pm$ 0.00003	5.41125 $\pm$ 0.00002
1 0 1	5.77683 $\pm$ 0.00007	5.77683 $\pm$ 0.00007	5.77660 $\pm$ 0.00007	5.77676 $\pm$ 0.00007	5.77722 $\pm$ 0.00007
1 0 2	7.4389 $\pm$ 0.0001	7.4395 $\pm$ 0.0001	7.4397 $\pm$ 0.0001	7.4401 $\pm$ 0.0001	7.4410 $\pm$ 0.0001
1 1 0	8.84832 $\pm$ 0.00003	8.84819 $\pm$ 0.00002	8.84796 $\pm$ 0.00002	8.84803 $\pm$ 0.00002	8.84833 $\pm$ 0.00002
1 0 3	9.5920 $\pm$ 0.0001	9.5925 $\pm$ 0.0001	9.59286 $\pm$ 0.00009	9.5941 $\pm$ 0.0001	9.59539 $\pm$ 0.00008
2 0 0	10.2212 $\pm$ 0.0001	10.22085 $\pm$ 0.00008	10.22052 $\pm$ 0.00007	10.22041 $\pm$ 0.00007	10.22080 $\pm$ 0.00007
1 1 2	10.37638 $\pm$ 0.00004	10.37645 $\pm$ 0.00004	10.37653 $\pm$ 0.00004	10.37685 $\pm$ 0.00004	10.37764 $\pm$ 0.00003
2 0 1	10.5736 $\pm$ 0.0002	10.5735 $\pm$ 0.0002	10.5737 $\pm$ 0.0001	10.5735 $\pm$ 0.0002	10.5739 $\pm$ 0.0001
2 0 2	11.5700 $\pm$ 0.0003	11.5698 $\pm$ 0.0003	11.5697 $\pm$ 0.0003	11.5701 $\pm$ 0.0003	11.5707 $\pm$ 0.0003
2 0 3	13.0642 $\pm$ 0.0002	13.0646 $\pm$ 0.0002	13.0645 $\pm$ 0.0002	13.0650 $\pm$ 0.0002	13.0667 $\pm$ 0.0002
2 1 0	13.5348 $\pm$ 0.0002	13.5344 $\pm$ 0.0001	13.5344 $\pm$ 0.0001	13.5341 $\pm$ 0.0001	13.5347 $\pm$ 0.0001
2 1 1	13.8041 $\pm$ 0.0003	13.8034 $\pm$ 0.0002	13.8033 $\pm$ 0.0002	13.8034 $\pm$ 0.0002	13.8037 $\pm$ 0.0002
1 0 5	14.4842 $\pm$ 0.0003	14.4853 $\pm$ 0.0003	14.4864 $\pm$ 0.0003	14.4881 $\pm$ 0.0003	14.4912 $\pm$ 0.0003
3 0 0	15.3568 $\pm$ 0.0002	15.3567 $\pm$ 0.0001	15.3565 $\pm$ 0.0001	15.3568 $\pm$ 0.0001	15.3572 $\pm$ 0.0001
2 1 3	15.8004 $\pm$ 0.0003	15.8005 $\pm$ 0.0003	15.8006 $\pm$ 0.0002	15.8017 $\pm$ 0.0002	15.8027 $\pm$ 0.0002
3 0 2	16.2911 $\pm$ 0.0002	16.2916 $\pm$ 0.0002	16.2916 $\pm$ 0.0002	16.2921 $\pm$ 0.0002	16.2929 $\pm$ 0.0002
2 2 0	17.7500 $\pm$ 0.0003	17.7505 $\pm$ 0.0003	17.7496 $\pm$ 0.0003	17.7501 $\pm$ 0.0002	17.7509 $\pm$ 0.0003

The tables report the peak positions, expressed as 2 θ , together with their uncertainties. Since the same reflections are present at all temperatures below the transition, there is no evidence of a structural phase change in this low-temperature region. The material keeps the β structure. After approximately 418 K, the diffractogram changes clearly. Some peaks that were present in the β phase disappear, and new peaks appear. This is the most important evidence from XRPD as it evidences the structural changes suffered by the crystal structure, due to the transformation into α -AgI, which has a cubic structure and thus a slightly different diffraction pattern.

This result agrees with the DSC analysis, where the phase transition was observed around 420 K. Therefore, XRPD confirms that the thermal event observed by DSC is not only a heat effect, but it is also connected with a real structural change in the material.

The cell parameters for β -AgI were calculated by peak refinement from the XRPD data using the tool WinPLOTR⁹ integrated in the software FullProf¹⁰. The parameters a and b are equal because the structure is hexagonal. The parameter c is different and must be analyzed separately (Table 5).

Table 5. Cell calculated for β phase of AgI parameters.

T / K	a = b / Å	c / Å	V / Å ³
294	4.59415 $\pm$ 0.00005	7.5130 $\pm$ 0.0002	137.318 $\pm$ 0.004
315	4.59409 $\pm$ 0.00005	7.5120 $\pm$ 0.0002	137.298 $\pm$ 0.004
335	4.59416 $\pm$ 0.00005	7.5110 $\pm$ 0.0002	137.291 $\pm$ 0.004
357	4.59415 $\pm$ 0.00005	7.5110 $\pm$ 0.0002	137.290 $\pm$ 0.004
375	4.59400 $\pm$ 0.00004	7.5100 $\pm$ 0.0002	137.258 $\pm$ 0.003

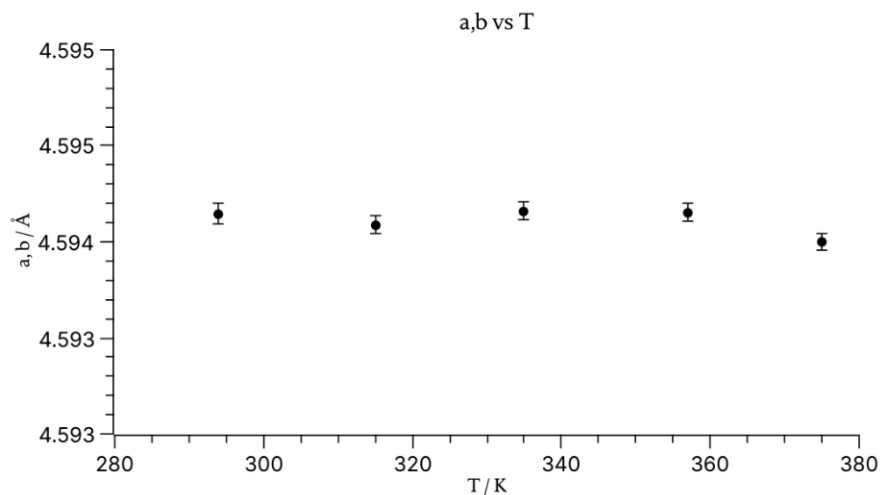


Figure 12. Cell parameter $a = b$ of β -Agl as a function of temperature.

The parameter $a = b$ remains almost constant between 294 K and 375 K (Figure 12). The changes are very small and close to the experimental uncertainty. This suggests that the basal plane of the hexagonal cell is not strongly affected by temperature before the transition.

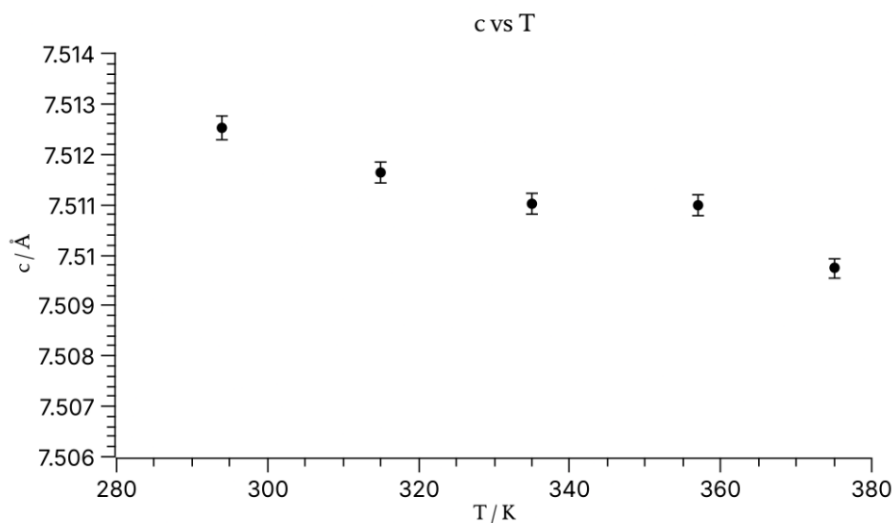


Figure 13. Cell parameter c of β -Agl as a function of temperature.

The parameter c shows a small decrease as temperature increases (Figure 13). This behavior indicates that the β phase does not expand uniformly in all crystallographic directions. In other words, the thermal response of the hexagonal cell is anisotropic. The structure is still stable, but the cell shrinks slightly along the c axis before the phase transition.

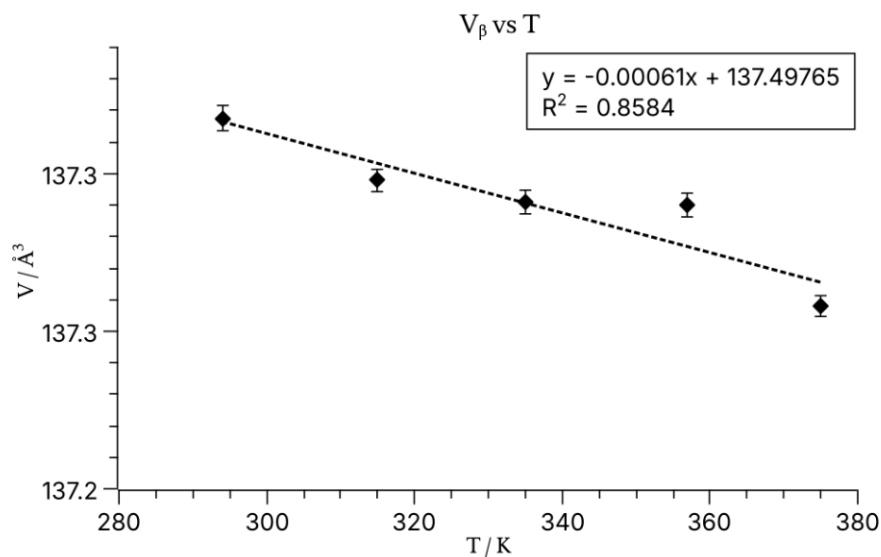


Figure 14. Unit-cell volume of β -AgI as a function of temperature.

The volume of the β phase decreases slightly with temperature before the transition (Figure 14). This behaviour can be interpreted as a case of negative thermal expansion, which in AgI has been related to the formation of Frenkel defects¹¹. Nevertheless, the change is small and should not be taken as evidence of a new phase. The main structural transformation only becomes clear when the diffraction pattern changes at higher temperatures.

3.3 Cell parameters of α -AgI

In a similar manner to β -AgI, the peaks from the diffraction pattern for α -AgI (Figure 15) were acquired at different temperatures (Table 6) and were refined using WinPLOTR.

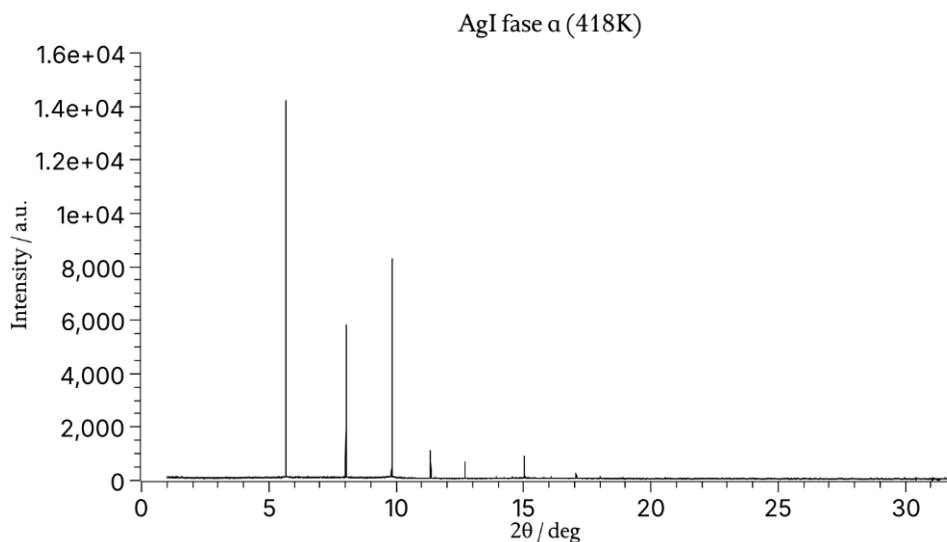


Figure 15. XRPD pattern of α -AgI at 418 K. The change in the diffraction peaks confirms the formation of the high-temperature cubic phase.

Table 6. XRPD diffraction peaks for α -phase of AgI.

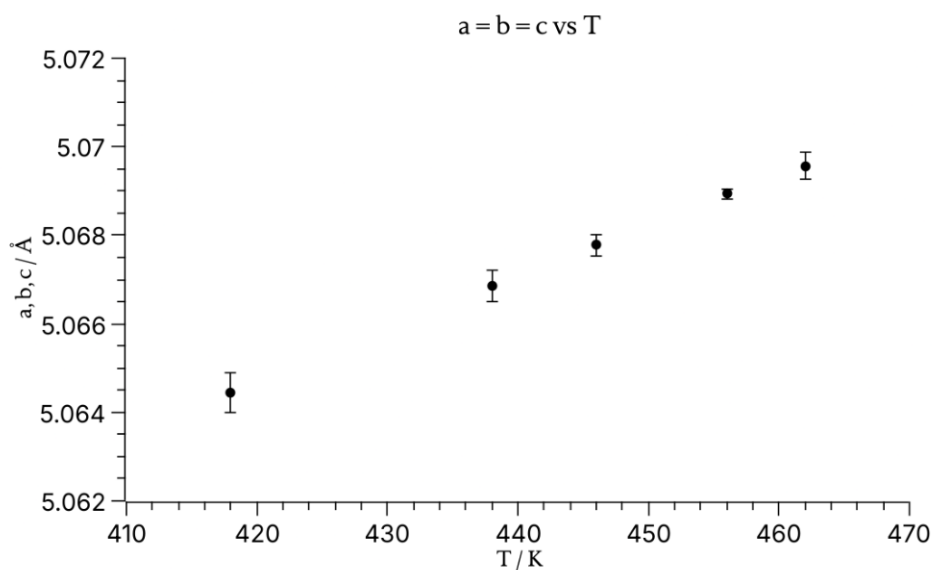
h k l	418 K 2 θ / deg	438 K 2 θ / deg	446 K 2 θ / deg	456 K 2 θ / deg	462 K 2 θ / deg
1 1 0	5.67229 $\pm$ 0.00001	5.66921 $\pm$ 0.00001	5.66801 $\pm$ 0.00001	5.66681 $\pm$ 0.00001	5.66576 $\pm$ 0.00002
2 0 0	8.02570 $\pm$ 0.00002	8.02131 $\pm$ 0.00002	8.01974 $\pm$ 0.00002	8.01788 $\pm$ 0.00002	8.01633 $\pm$ 0.00003
2 1 1	9.83323 $\pm$ 0.00002	9.82782 $\pm$ 0.00002	9.82619 $\pm$ 0.00002	9.82385 $\pm$ 0.00002	9.82218 $\pm$ 0.00002
2 2 0	11.35930 $\pm$ 0.00005	11.35297 $\pm$ 0.00005	11.35094 $\pm$ 0.00005	11.34824 $\pm$ 0.00005	11.34671 $\pm$ 0.00005
3 1 0	12.70564 $\pm$ 0.00007	12.69881 $\pm$ 0.00007	12.69592 $\pm$ 0.00007	12.69280 $\pm$ 0.00007	12.69099 $\pm$ 0.00008
2 2 2	13.9235 $\pm$ 0.0002	13.9164 $\pm$ 0.0002	13.9133 $\pm$ 0.0003	13.9102 $\pm$ 0.0003	13.9085 $\pm$ 0.0003
3 2 1	15.04526 $\pm$ 0.00006	15.03726 $\pm$ 0.00006	15.03433 $\pm$ 0.00006	15.03081 $\pm$ 0.00007	15.02903 $\pm$ 0.00007
4 0 0	16.0898 $\pm$ 0.0003	16.0819 $\pm$ 0.0003	16.0794 $\pm$ 0.0003	16.0755 $\pm$ 0.0003	16.0727 $\pm$ 0.0004
3 3 0	17.0733 $\pm$ 0.0002	17.0644 $\pm$ 0.0002	17.0611 $\pm$ 0.0002	17.0575 $\pm$ 0.0002	17.0548 $\pm$ 0.0002

Table 7. Cell parameters calculated for the α phase of AgI.

T / K	a / Å	V / Å ³
418	5.0645 $\pm$ 0.0005	129.90 $\pm$ 0.04
438	5.0669 $\pm$ 0.0004	130.08 $\pm$ 0.03
446	5.0678 $\pm$ 0.0002	130.15 $\pm$ 0.02
456	5.0689 $\pm$ 0.0001	130.242 $\pm$ 0.009
462	5.0696 $\pm$ 0.0003	130.29 $\pm$ 0.02

Above the transition temperature, the diffraction peaks are indexed as α -AgI. This phase has a cubic structure, so the three cell parameters are equal (Table 7):

$$a = b = c$$

Figure 16. Cell parameters $a = b = c$ of α -AgI as a function of temperature.

For the α phase, the cell parameter increases with temperature (Figures 16-17). This behaviour is expected for a high-temperature phase because the atoms vibrate anharmonically, increasing the average interatomic distances and causing the unit cell to expand.

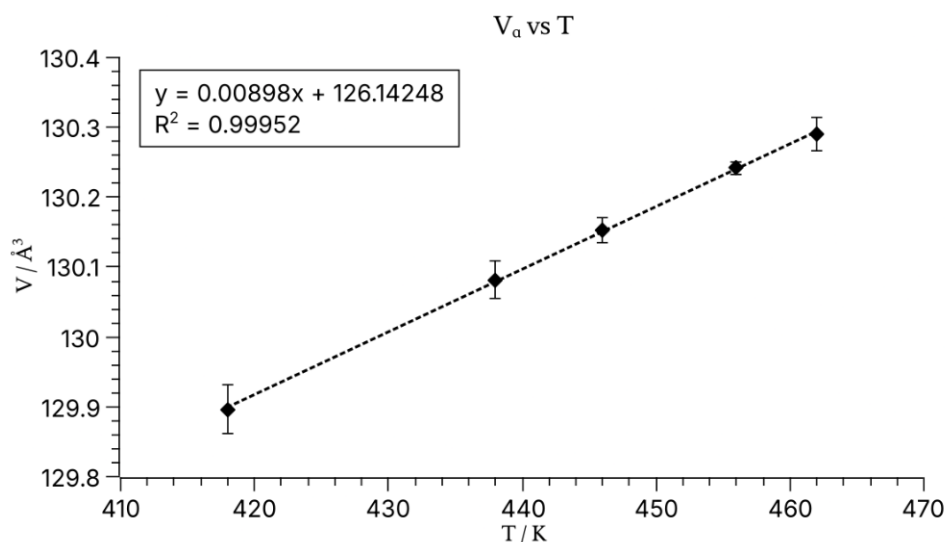


Figure 17. Unit-cell volume of α -AgI as a function of temperature.

The volume of α -AgI increases almost linearly with temperature (Figure 17). The linear fit has a very good agreement with the experimental points, which means that the expansion of the cubic phase is regular in this temperature range. This is different from the β phase, where the variation was smaller and not equally distributed in all directions.

3.4 Comparison between β -AgI and α -AgI

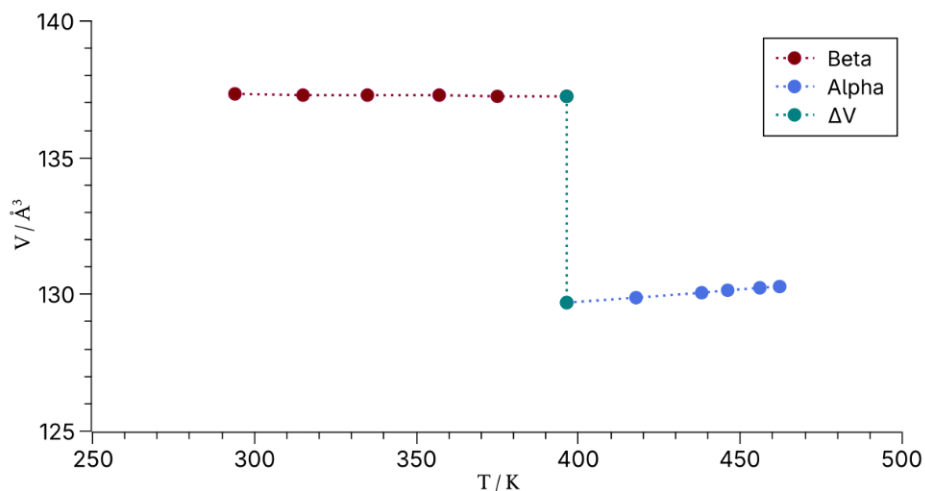


Figure 18. Comparison of the unit-cell volume (V) of β -AgI and α -AgI as a function of temperature (T). The vertical line shows the volume change associated with the phase transition.

When both phases are compared, the transition becomes evident. The β phase is present at low temperature and has a larger refined unit-cell volume. The α phase appears above the transition

and shows a smaller unit-cell volume, followed by a regular increase with temperature (Figure 18).

The volume change between the two phases cannot be explained only as a normal thermal expansion effect. It is mainly associated with the $\alpha \rightarrow \beta$ phase transition of AgI, in which the structure changes from the hexagonal β phase to the cubic α phase.

The XRPD results confirm the phase transition of AgI around 418-420 K. Below this temperature, the diffraction patterns and the cell parameters are consistent with β -AgI. Above this temperature, the appearance of new peaks and the cubic cell parameters confirm the formation of α -AgI. This supports the DSC results and gives direct structural evidence of the phase transition.

4. UV-Visible Diffuse Reflectance Spectroscopy (UV/Vis-DRS)

4.1 Principle of the Technique

UV-Vis diffuse reflectance spectroscopy (UV/Vis-DRS) measures the fraction of incident light reflected by a powdered sample as a function of the wavelength from the light source. For a semiconducting or insulating solid, the spectrum shows high reflectance at long wavelengths and a sharp drop in absorption at the wavelength corresponding to the optical band gap, where photons begin to have enough energy to promote electrons across the gap¹². To extract quantitative optical information from reflectance data, the Kubelka-Munk transformation, $F(R_\infty) = (1 - R_\infty)^2 / 2R_\infty$, is applied to convert reflectance (R_∞) into a quantity proportional to the absorption coefficient. The optical band gap (E_g) is then obtained from a Tauc plot, in which (edge to the energy axis gives E_g ($n = 2$ was used here, appropriate for a direct allowed transition). $F(R_\infty) \cdot E^n$ is plotted against photon energy E ; extrapolating the linear region of the absorption.

4.2 Optical Response and Band-Gap Estimation of AgI

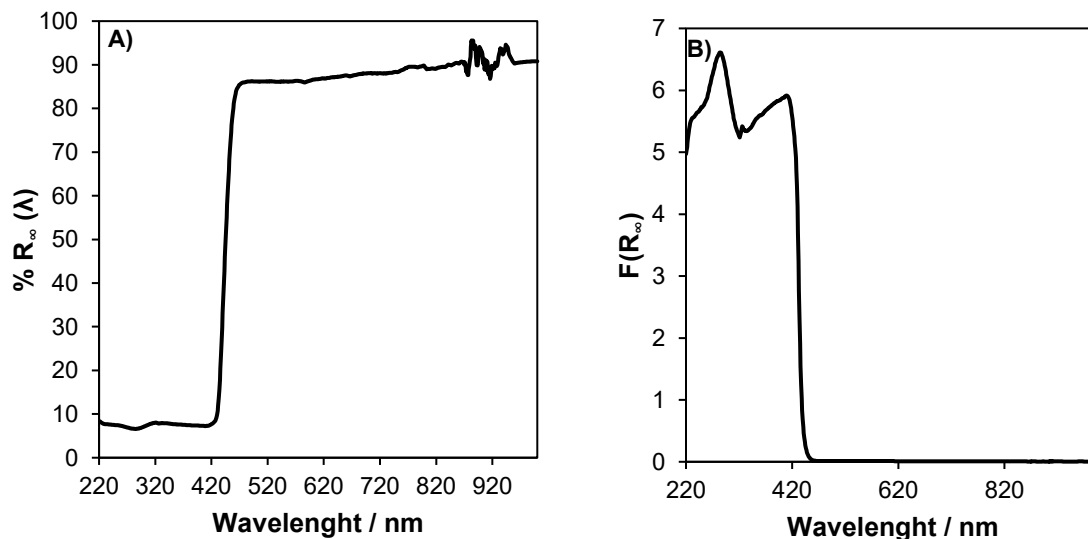


Figure 19. (A) UV/Vis-DRS spectrum of AgI. (B) Corresponding Kubelka-Munk plot of AgI.

The diffuse reflectance spectrum of AgI (Figure 19a) shows low reflectance below ~420 nm and a very sharp rise to high reflectance (~85-90%) above this wavelength, indicating a well-defined, steep absorption edge typical of a relatively narrow-gap, highly crystalline semiconductor. The

corresponding Kubelka-Munk plot (Figure 19b) shows a strong, broad absorption band below ~420 nm with a maximum $F(R_\infty)$ around 6-7, consistent with the much higher intrinsic absorption of AgI.

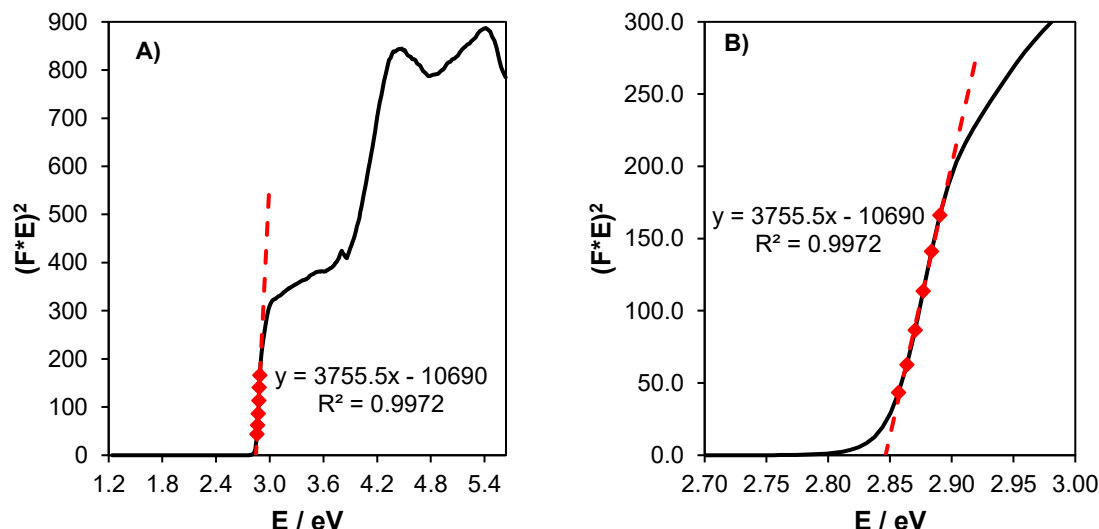


Figure 20. (A) Tauc plot, $(F(R_\infty) \cdot E)^2$, for AgI over the full measured energy range. (B) Magnified view of the Tauc plot for estimating the band gap of AgI.

Table 8. Band gap calculation from Tauc plot for AgI.

Parameter						E_g
m	δm	R^2	q	δq	δy	(eV)
3755.5	99.5	0.9972	-10689.8	285.9	2.7718	2.8 ± 0.1

Extrapolation of the linear region of the Tauc plot to the energy axis (Figure 20), gives an optical band gap for AgI of 2.8 ± 0.1 eV (Table 8), in reasonable agreement with literature value of 2.58 eV¹³ for the direct optical gap of AgI in its low-temperature β phase. The corresponding wavelength is $\lambda = hc/E(\text{eV}) = 1240/2.8 \approx 443$ nm (visible violet/blue) This is why AgI appears yellowish as it absorbs blue/violet light and reflects yellow and red.

This optical estimate is distinct from and, as discussed in Section 5, larger than the small thermal activation energies obtained from the EIS conductivity analysis, because the optical gap reflects direct electronic excitation across the full band gap, whereas the EIS activation energies reflect the much smaller energies associated with ionic hopping and electronic carrier excitation from shallow defect levels.

5. Electrochemical Impedance Spectroscopy (EIS) of AgI

5.1 Principle of the Technique

- Electrochemical Impedance Spectroscopy (EIS) measures the electrical impedance, which can be approximated through the sample resistance of a material as a function of temperature, allowing the electrical conductivity (σ) to be determined from the sample's

geometry or cell parameter, and resistance. By measuring conductivity over a range of temperatures, an Arrhenius-type analysis can be performed to extract the activation energy for charge transport¹⁴. Two transport models were considered for AgI: an electronic conductivity model, in which $\ln(\sigma)$ is plotted against $1/T$, and an ionic conductivity model, in which $\ln(T\sigma)$ is plotted against $1/T$, in which the additional temperature factor accounts for the temperature dependence of the pre-exponential term expected for thermally activated ionic hopping. In the former case the activation energy is obtained from the slope (m) of the linear Arrhenius fit via $E_g = -2 \cdot m \cdot k_B$; while on the later case the slope (m) is related to $E + \Delta F_H/2 = -m \cdot k_B$.

5.2 Oven Temperature Calibration

Table 9. Oven temperature verification with respect to reference thermocouple temperature.

Oven Temp. (°C)	Oven Temp. (K)	Thermocouple T. (°C)	Thermocouple T. (K)
24	297.15	25.4	298.55
35	308.15	40.0	313.15
50	323.15	59.5	332.65
65	338.15	78.6	351.75
80	353.15	97.9	371.05
95	368.15	118.3	391.45
110	383.15	137.1	410.25
125	398.15	154.8	427.95
140	413.15	172.5	445.65
155	428.15	190.7	463.85
170	443.15	207.8	480.95
185	458.15	224.5	497.65
200	473.15	242.4	515.55

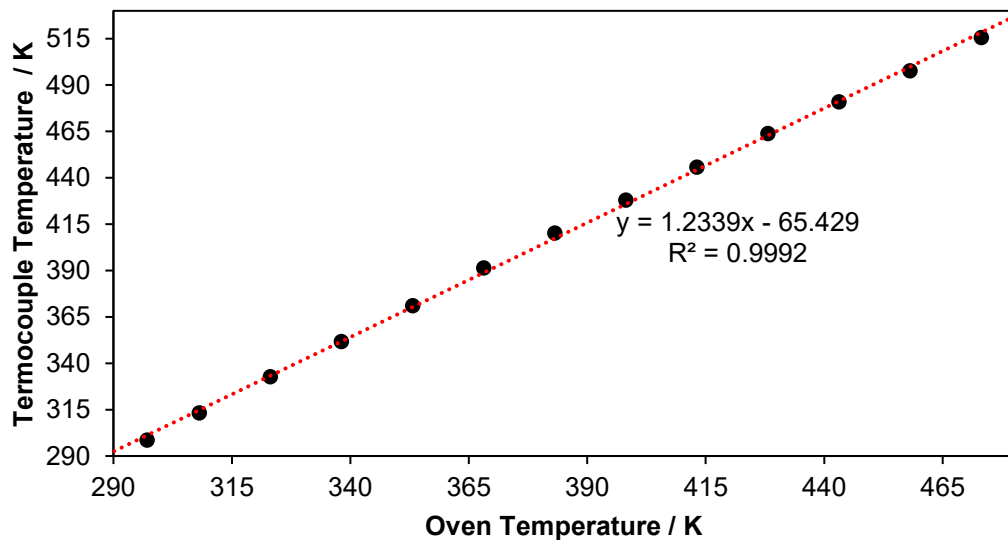


Figure 21. Thermocouple temperature vs. oven (set-point) temperature.

Due to the sensibility required for the temperature measurements when EIS experiments are carried out, a verification of the oven temperature was performed with a thermocouple to correct the readings at the different stages of the experiment (Table 9). The thermocouple readings resulted systematically higher than the oven registered temperatures, hence the linear regression model from figure 21 was used to correct all subsequent measurements.

5.3 Resistance and Conductivity Measurements

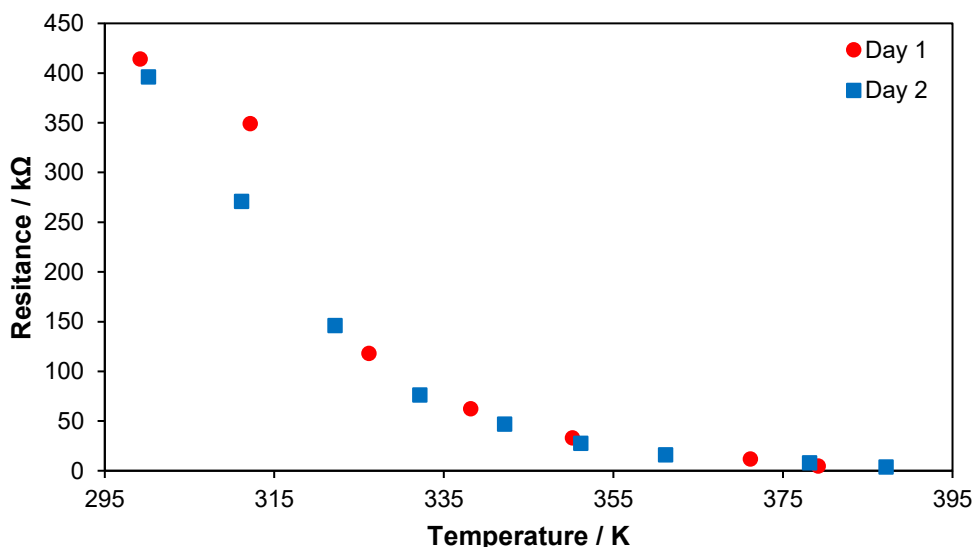


Figure 22. Resistance of the AgI cell as a function of temperature, corrected measurements carried out on two separate days.

The resistance of the prepared AgI pellet (Table 10) decreases non-linearly with increasing temperature, as can be seen on the Nyquist plot (Figure 22-23), from several hundred kΩ near room temperature to a few kΩ above ~370 K, consistent with thermally activated conduction. The close agreement between day 1 and day 2 datasets indicates good measurement reproducibility and sample stability over the timescale of the experiment.

Table 10. Cell geometry parameters used to convert resistance into conductivity for the AgI sample.

Parameter	Value
Width (cm)	0.12
Diameter (cm)	1.32
Cell parameter (m^{-1})	8.77

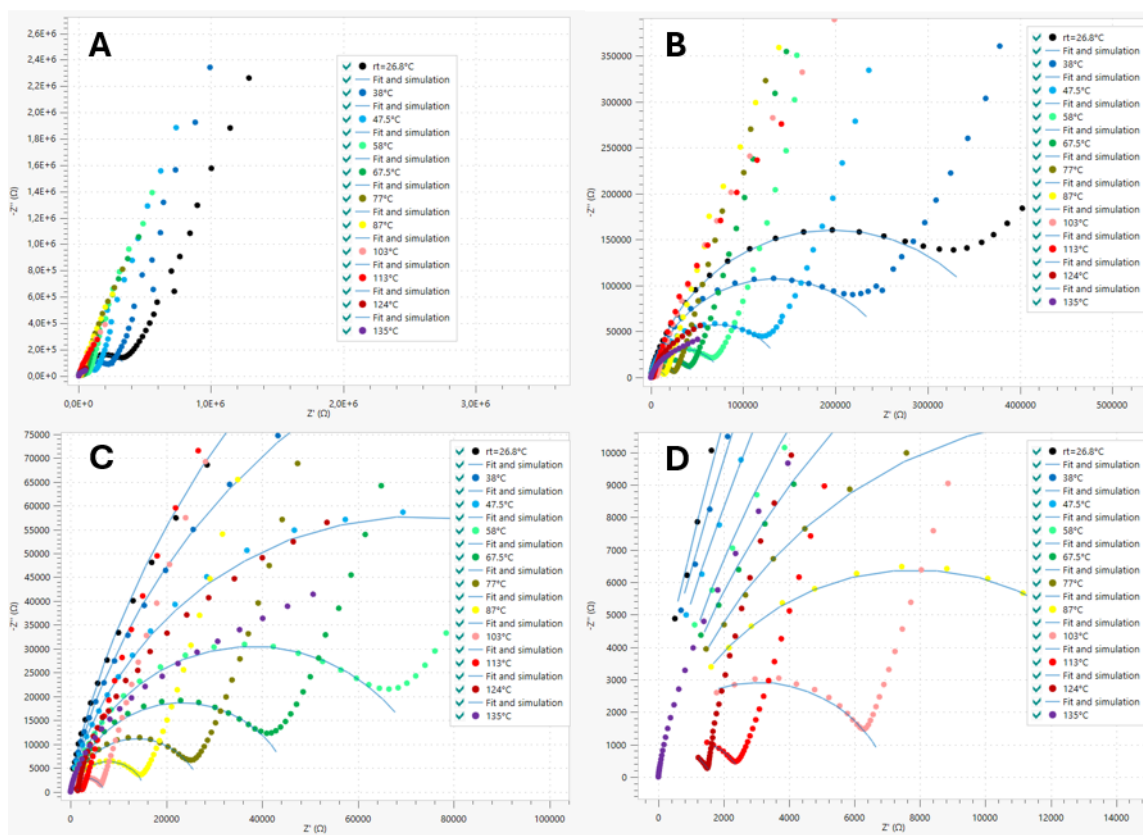


Figure 23. Nyquist plots portraying the resistance measurements of AgI pellet sample by EIS at different temperatures. Images A-D display different zoom levels for better clarity of data obtained at higher temperatures.

Table 11. Electronic Conductivity Model: Day 1

Oven T (K)	Corrected T (K)	T^{-1} (K $^{-1}$)	R (kΩ)	σ (mS m $^{-1}$)	ln(σ)
299.15	303.70	0.00329	414	0.0212	-3.8547
312.15	319.74	0.00313	349	0.0251	-3.6839
326.15	337.01	0.00297	118	0.0743	-2.5995
338.15	351.82	0.00284	62.3	0.1408	-1.9608
350.15	366.63	0.00273	33	0.2657	-1.3253
371.15	392.54	0.00255	11.7	0.7495	-0.2884
379.15	402.41	0.00249	4.79	1.8307	0.6047
401.15	429.56	0.00233	N.F.P. ^a	—	—
409.15	439.43	0.00228	N.F.P. ^a	—	—

^a Not fittable point

Table 12. Electronic Conductivity Model: Day 2

Oven T (K)	Corrected T (K)	T ⁻¹ (K ⁻¹)	R (kΩ)	σ (mS m ⁻¹)	ln(σ)
300.15	304.93	0.00328	396	0.0221	-3.8102
311.15	318.51	0.00314	271	0.0324	-3.4309
322.15	332.08	0.00301	146	0.0601	-2.8124
332.15	344.42	0.00290	76.2	0.1151	-2.1622
342.15	356.76	0.00280	46.7	0.1878	-1.6725
351.15	367.86	0.00272	27.4	0.3200	-1.1393
361.15	380.20	0.00263	15.8	0.5550	-0.5888
378.15	401.18	0.00249	8.04	1.0907	0.0868
387.15	412.28	0.00243	3.85	2.2776	0.8231
401.15	429.56	0.00233	N.F.P. ^a	—	—
409.15	439.43	0.00228	N.F.P. ^a	—	—

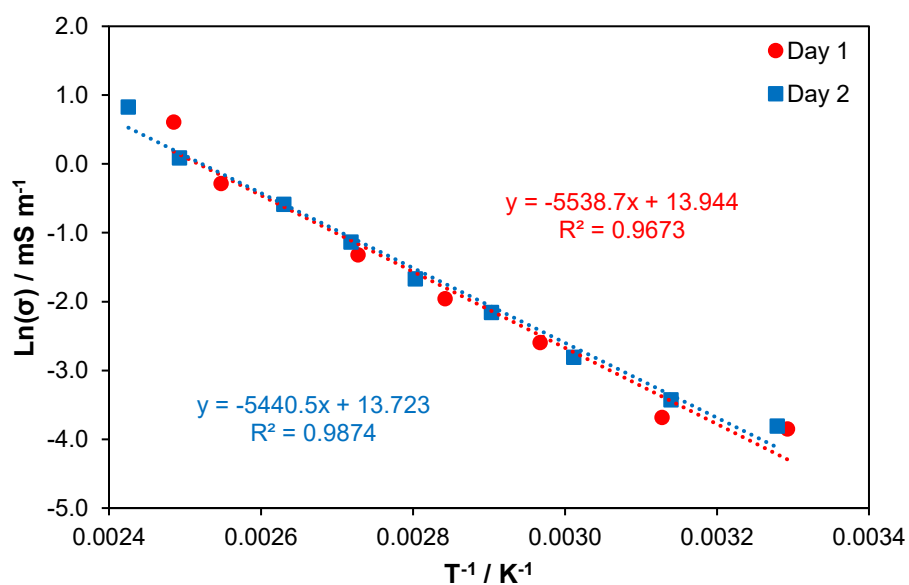
^a Not fittable point

Figure 24. Arrhenius plot, ln(σ) vs. 1/T, for AgI on Day 1 and Day 2, analyzed using the electronic conductivity model.

Table 13. Band gap energy calculated for AgI based on electronic conductivity model fitting.

Day	Parameter						Band Gap Energy	
	m	δm	R ²	q	δq	δy	E _g = -2mk _B (J)	E _g (eV)
1	-5538.7	455.2	0.9673	13.944	1.306	0.3302	(1.5 ± 0.1) × 10 ⁻¹⁹	0.95 ± 0.08
2	-5440.5	232.2	0.9874	13.723	0.658	0.1895	(1.50 ± 0.06) × 10 ⁻¹⁹	0.94 ± 0.04

As can be seen from the data portrayed in table 12 and 13, as well as in figure 24, both datasets show excellent linear Arrhenius behavior over the full measured temperature range, with the two independent measurement days giving statistically consistent activation energies of 0.95 eV and 0.94 eV for day 1 and day 2, respectively (Table 13). However, in order to confirm if the conduction model for the AgI system is mainly electronic as this data seems to suggest, we performed an alternative analysis of the same dataset that considers the diffusion of Ag vacancies and Frenkel defects, resembling more an ionic conduction model (Table 14-15). For the electronic conduction model to prevail, it should agree with the ionic conduction model, otherwise it can be disproved.

Table 14. Ionic Conductivity Model: Day 1

Corrected T (K)	T^{-1} (K^{-1})	R ($k\Omega$)	σ ($mS\ m^{-1}$)	$\ln(T\sigma)$
303.70	0.00329	414	0.0212	1.8614
319.74	0.00313	349	0.0251	2.0836
337.01	0.00297	118	0.0743	3.2206
351.82	0.00284	62.3	0.1408	3.9024
366.63	0.00273	33	0.2657	4.5790
392.54	0.00255	11.7	0.7495	5.6843
402.41	0.00249	4.79	1.8307	6.6022
429.56	0.00233	N.F.P. ^a	—	—
439.43	0.00228	N.F.P. ^a	—	—

^a Not fittable point

Table 15. Ionic Conductivity Model: Day 2

Corrected T (K)	T^{-1} (K^{-1})	R ($k\Omega$)	σ ($mS\ m^{-1}$)	$\ln(T\sigma)$
304.93	0.00328	396	0.0221	1.9099
318.51	0.00314	271	0.0324	2.3327
332.08	0.00301	146	0.0601	2.9930
344.42	0.00290	76.2	0.1151	3.6797
356.76	0.00280	46.7	0.1878	4.2045
367.86	0.00272	27.4	0.3200	4.7684
380.20	0.00263	15.8	0.5550	5.3519
401.18	0.00249	8.04	1.0907	6.0812
412.28	0.00243	3.85	2.2776	6.8448
429.56	0.00233	N.F.P. ^a	—	—
439.43	0.00228	N.F.P. ^a	—	—

^a Not fittable point

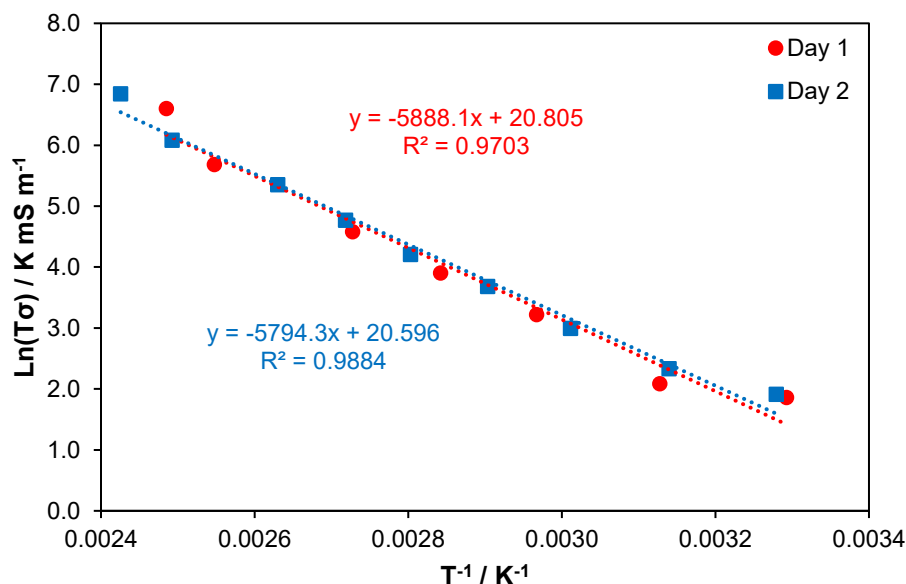


Figure 25. Arrhenius plot, $\ln(T\sigma)$ vs. $1/T$, for AgI on Day 1 and Day 2, analyzed using the ionic conductivity model.

Table 16. Band gap energy calculated for AgI based on ionic conductivity model fitting.

Day	Parameter						Band Gap Energy	
	m	δm	R^2	q	δq	δy	$E + \delta H/2 = -mk_B$ (J)	$E + \delta H/2$ (eV)
1	-5888.1	460.4	0.9703	20.805	1.321	0.3340	$(8.1 \pm 0.6) \times 10^{-20}$	0.51 ± 0.04
2	-5794.3	237.3	0.9884	20.596	0.673	0.1937	$(8.0 \pm 0.3) \times 10^{-20}$	0.50 ± 0.02

According to Einstein-Smoluchowski the relationship between conductivity and ionic diffusion is as follows: $\sigma = nq^2D/k_BT$, Where D is the diffusion coefficient (which has an Arrhenius form). So:

$$T\sigma \propto \exp[-E/k_BT]$$

The Arrhenius plot (Figure 25) give very similar energy values for the two measurement days, 0.51 eV and 0.50 eV, showing good reproducibility (Table 16). However, this does not support the electronic conduction model. For electronic conduction, twice this value should be comparable to the band gap, giving around 1.0 eV. This is far from the band gap obtained from the UV-Vis data, $E_g \approx 2.8$ eV. Therefore, the slope of the Arrhenius plot is not related to the band gap, but rather to the ionic conduction process, where the measured energy includes the activation energy and part of the defect formation enthalpy, approximately $E_a + (\delta H/2)$.

The EIS measurements show that the electrical resistance of AgI decreases by more than two orders of magnitude between room temperature and ~ 400 K, accurately described by Arrhenius-type thermally activated conduction in both analysis models, with good day-to-day reproducibility. All EIS measurements were carried out below the $\alpha \leftrightarrow \beta$ transition identified by DSC (Section 2, ~ 411 – 422 K), so the resistance and conductivity trends reported here are characteristic of the lower-temperature β -phase of AgI rather than the highly conductive superionic α -phase, which forms above the transition temperature.

6. Overall Conclusion

The AgI part shows coherent behavior between the different techniques. DSC clearly shows the reversible $\beta \leftrightarrow \alpha$ transition close to 420 K, with the expected hysteresis between heating and cooling. According to the Ehrenfest classification, this transition can be classified as a first-order phase transition, because it involves discontinuities in the first derivatives of the Gibbs free energy with respect to temperature and pressure, namely entropy and volume¹⁵. This is consistent with the DSC signal, where the endothermic discontinuity in the HF for the $\beta \rightarrow \alpha$ transition, and exothermic discontinuity for the $\alpha \rightarrow \beta$ transition is observed, as well as with the XRPD results, which confirm that this thermal event is also a real structural change from the low-temperature hexagonal β -phase to the high-temperature cubic α -phase. The cell parameters before the transition change only slightly, but after the transition the diffraction pattern changes strongly, supporting the first-order character of the transformation. UV-Vis gives an optical band gap around 2.8 eV, showing that AgI behaves as a narrow-gap semiconductor, although this value should not be confused with the activation energies obtained from EIS, since they describe different physical processes.

The EIS measurements complete the picture by showing that the resistance decreases strongly with temperature, so the electrical transport is thermally activated. The activation energy obtained with the ionic model is lower than with the electronic model, which makes sense for AgI because its most important conduction mechanism is related to silver ion mobility. However, since the conductivity measurements were mainly performed below the $\beta \rightarrow \alpha$ transition, they mostly describe the β -phase and not the fully superionic α -phase. In general, the results agree well: DSC detects the transition, XRPD proves the structural change, UV-Vis describes the optical response, and EIS shows how transport increases as the material approaches the transition region.

Introduction

Cerium oxide (CeO₂) is a very common oxide in solid-state chemistry because it has a stable fluorite-type structure. In this structure, the Ce⁴⁺ ions form a cubic framework and the oxygen ions occupy the spaces inside it (Figure 26a). A simple way to imagine it is that the structure is quite ordered and resistant: pure CeO₂ does not show, under normal conditions, a phase transition like AgI. For this reason, in CeO₂ the interesting part is not a thermal transformation, but what happens when the lattice is modified by doping. When part of the Ce⁴⁺ ions are replaced by Gd³⁺ ions from a metal oxide like Gd₂O₃ (Figure 26b), the crystal has to compensate the difference in charge and size. The most usual way to do this is by creating oxygen vacancies: for every two Gd³⁺ ions introduced, one oxygen vacancy is formed. These vacancies are important because they can help oxygen ions to move through the structure, which is why Gd-doped ceria (Figure 26c) is widely studied as an oxygen-ion conductor, especially for solid oxide fuel cells and related electrochemical devices. In addition, considering that Gd³⁺ ions are larger than Ce⁴⁺ ions, the unit cell is expected to expand when the Gd content increases. This is a simple but important sign that Gd is entering the CeO₂ lattice and not only staying as a separated phase¹⁶.

In this work, Gd-doped CeO₂ samples of the general formula Ce_{1-x}Gd_xO_{2-(x/2)}, with different nominal compositions, $x_{\text{Gd}} = 0, 0.125, 0.250, 0.375$ and 0.500 , were prepared by a Pechini-type sol-gel route (Figure 27). In this method, the metal ions are first mixed in solution and then trapped inside an organic polymer-like network, formed mainly with citric acid and ethylene glycol. This helps to keep Ce and Gd more homogeneously distributed before calcination, instead of starting from a simple mechanical mixture of powders. Then, the samples were studied using scanning electron microscopy (SEM) and transmission electron microscopy (TEM) to observe morphology, elemental composition, and particle size, XRPD to follow the crystal structure and cell parameters, UV-Vis to estimate the optical band gap, and Raman spectroscopy to observe local disorder and defect-related changes. In this way, the section connects the Gd content with the structural, morphological and optical behaviour of CeO₂¹⁷.

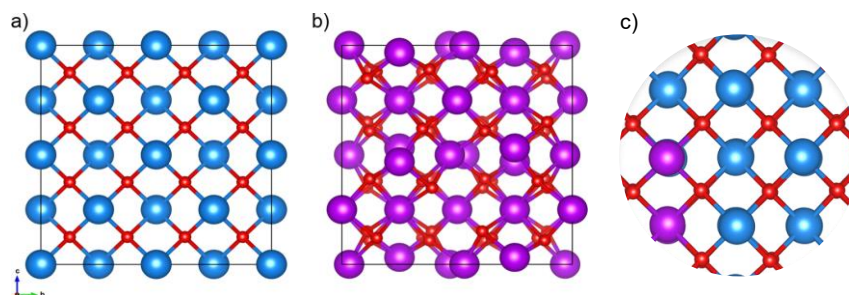


Figure 26. CeO₂ (a), Gd₂O₃ (b) and Ce_{1-x}Gd_xO_{2-(x/2)} (c) supercell crystal structure in bulk¹⁸.

1. Synthesis of Gd-Doped CeO₂ by the Pechini Sol-Gel Method

Gd-doped CeO₂ samples were prepared by a Pechini-type sol-gel technique, in which Gd³⁺ is incorporated into the CeO₂ fluorite lattice to create oxygen vacancies and achieve atomic-level doping. The sol-gel route was chosen because it allows the Ce⁴⁺ and Gd³⁺ precursor ions to mix uniformly at the atomic scale while still in the liquid (sol) state, before solidification, avoiding the segregation that can occur with solid-state mixing methods.

CeO₂ adopts the fluorite-type structure, in which Ce⁴⁺ is in cubic (eight-fold) coordination with oxygen. Because the ionic radius of Gd³⁺ of approximately 1.05 Å is larger than that of Ce⁴⁺, ≈0.97 Å, the substitution of Gd³⁺ into the Ce⁴⁺ sites has some consequences.

First, the charge compensation. Two Gd³⁺ ions substituting for two Ce⁴⁺ ions leave the lattice with a net negative effective charge, which is compensated by the formation of one oxygen vacancy per two dopant ions. These oxygen vacancies can also act as F-centers that trap electrons and absorb visible light, producing the yellow-brown coloration typically observed in these materials, the higher the vacancy concentration, the darker the color (Figure 27A). Then, there is the ionic transport. Here the oxygen vacancies created by aliovalent doping provide hopping sites for oxygen ions, which is the structural basis for using doped CeO₂ as an oxygen-ion (solid-electrolyte) conductor.

In the synthesis, citric acid acts as a chelating agent: its three carboxylic groups and one hydroxyl group coordinate to the Gd³⁺ and Ce⁴⁺ ions, preventing premature precipitation. Ethylene glycol acts as a cross-linking agent, undergoing polyesterification with citric acid to form a three-dimensional polymeric network that traps the metal ions homogeneously. On calcination, thermal decomposition of the residual nitrate ions provides an internal oxidizing environment that assists removal of the organic matrix and crystallization of the doped oxide (Figure 27B).

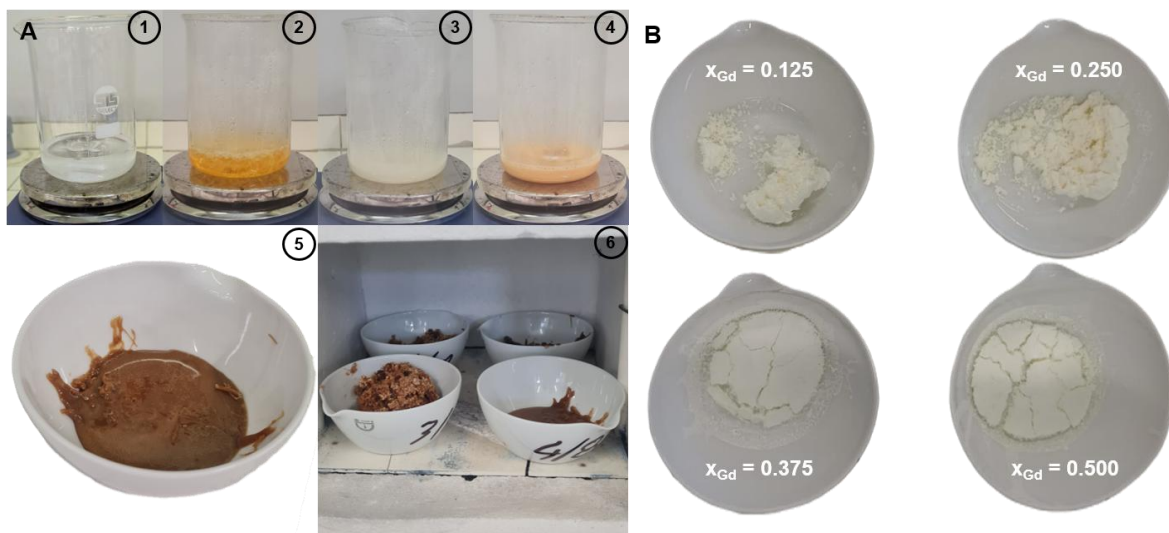


Figure 27. A) Physical changes of Ce_{1-x}Gd_xO_{2-(x/2)} samples during sol-gel synthesis process before firing, and B) products of synthesis after heat treatment at 500 °C for 5 h.

2. Scanning Electron Microscopy and Electron Probe Microanalysis (SEM/EMPA)

2.1 Principle of the Technique

SEM/EMPA was used to study the morphology and chemical composition of the Ce_{1-x}Gd_xO_{2-(x/2)} powders. In SEM, a focused electron beam scans the surface of the sample and the emitted electrons are used to form an image, allowing to observe the general shape, texture, porosity and agglomeration of the particles. This is useful because the incorporation of Gd can change the way the grains grow and group together after synthesis. EMPA gives elemental information from the characteristic X-rays emitted by the sample. This helps to obtain the elemental composition of Ce and Gd content present the samples, thus verifying if the proportions of the synthesized samples are close to the expected composition. Therefore, this technique does not only show how the

powder looks, but also gives an idea of whether the dopant was effectively incorporated in the analyzed samples¹⁹.

2.2 Chemical Composition and Morphological Analysis

Table 17. Elemental composition analysis by SEM/EMPA of the synthesized $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ samples.

Sample $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$	Physical state	Rep.	Atomic % Ce	Mean % Ce	Atomic % Gd	Mean % Gd	x_{Gd} Theoric	x_{Gd} Real
$\text{Ce}_{0.875}\text{Gd}_{0.125}\text{O}_{1.938}$	Powder	1	34.75	34.81 ± 0.08	5.25	5.19 ± 0.08	0.125	0.130 ± 0.002
		2	34.93		5.07			
		3	34.75		5.25			
		4	34.84		5.16			
		5	34.79		5.21			
	Pellet	1	34.70	35.1 ± 0.3	5.30	4.9 ± 0.3	0.125	0.123 ± 0.007
		2	35.31		4.69			
		3	35.40		4.60			
		4	34.86		5.14			
		5	34.97		5.03			
		6	35.26		4.74			
$\text{Ce}_{0.750}\text{Gd}_{0.250}\text{O}_{1.875}$	Powder	1	30.21	30.2 ± 0.2	9.79	9.8 ± 0.2	0.250	0.244 ± 0.005
		2	30.08		9.92			
		3	30.03		9.97			
		4	30.20		9.80			
		5	30.29		9.71			
		6	30.54		9.46			
	Pellet	1	30.18	30.4 ± 0.2	9.82	9.6 ± 0.2	0.250	0.241 ± 0.005
		2	30.48		9.52			
		3	30.29		9.71			
		4	30.57		9.43			
$\text{Ce}_{0.625}\text{Gd}_{0.375}\text{O}_{1.813}$	Powder	1	24.75	24.7 ± 0.3	15.25	15.3 ± 0.3	0.375	0.383 ± 0.007
		2	24.75		15.25			
		3	24.37		15.63			
		4	24.39		15.61			
		5	24.90		15.10			
		6	25.00		15.00			
	Pellet	1	25.25	25.4 ± 0.2	14.75	14.6 ± 0.2	0.375	0.365 ± 0.004
		2	25.30		14.70			
		3	25.67		14.33			
		4	25.38		14.62			
		5	25.43		14.57			
		6	25.31		14.69			
$\text{Ce}_{0.500}\text{Gd}_{0.500}\text{O}_{1.750}$	Powder	1	20.23	20.1 ± 0.1	19.77	19.9 ± 0.1	0.500	0.498 ± 0.004
		2	20.23		19.77			
		3	20.06		19.94			
		4	20.02		19.98			
		5	20.03		19.97			
		6	19.90		20.10			
	Pellet	1	20.56	20.7 ± 0.1	19.44	19.3 ± 0.1	0.500	0.483 ± 0.003
		2	20.68		19.32			
		3	20.68		19.32			
		4	20.61		19.39			
		5	20.84		19.16			
		6	20.70		19.30			

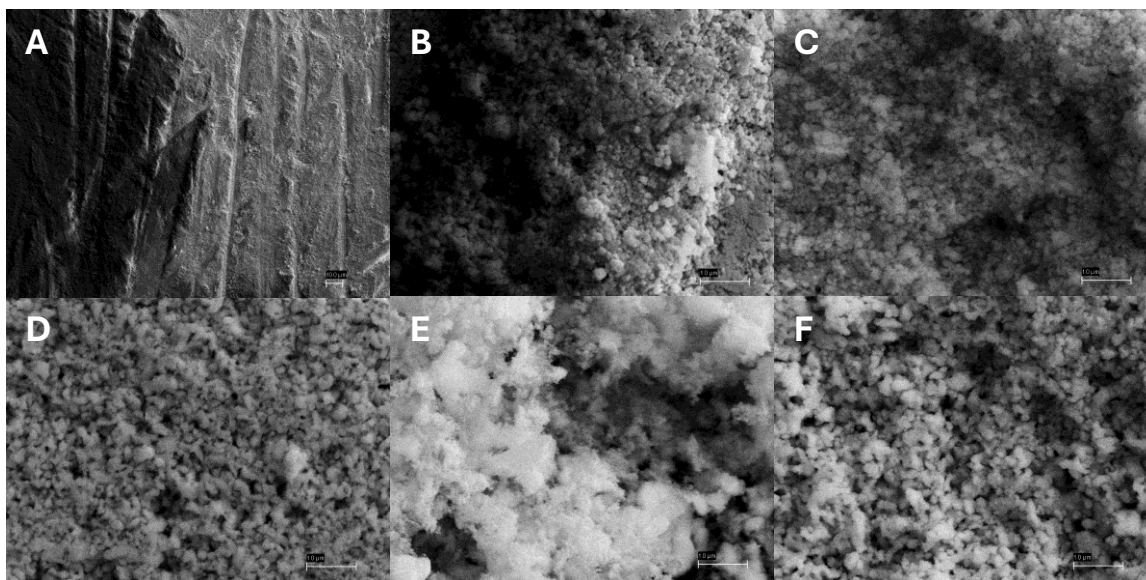


Figure 28. SEM images of the morphological changes of the unaltered CeO_2 (A-B) with respect to $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ composites doped with increasing concentrations of Gd after heat treatment at 500 °C for 5 h: C) $x_{\text{Gd}}=0.130$. D) $x_{\text{Gd}}=0.244$; E) $x_{\text{Gd}}=0.383$; F) $x_{\text{Gd}}=0.498$.

The SEM images show the morphological evolution of CeO_2 as the Gd content increases (Table 17, Figure 28). In the undoped CeO_2 sample, shown in Images A and B, the material presents a relatively compact, dense, and continuous morphology, with a surface that appears smoother and more bulk-like compared with the doped samples. When gadolinium is introduced at low to moderate concentrations, as observed in Images C and D, the morphology changes noticeably toward a more granular and particulate texture. In these images, distinct grains and clusters become more evident, which may be associated with lattice strain, defect formation, and the inhibition of bulk crystal growth caused by Gd incorporation. At the highest Gd concentration, shown in Images E and F, the sample appears more porous, irregular, and strongly agglomerated, with smaller particles forming rough “cauliflower-like” clusters. Overall, the SEM analysis indicates a clear transition from a compact bulk-like morphology in pure CeO_2 to a more dispersed, granular, porous, and agglomerated structure as the Gd fraction increases. This increase in roughness and porosity may also suggest a higher specific surface area, which is relevant for applications such as solid oxide fuel cells or catalysis.

3. Transmission Electron Microscopy (TEM)

3.1 Principle of the Technique

TEM was used to observe the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ nanoparticles at higher resolution and to estimate their average particle size. In this technique, an electron beam passes through very thin regions of the sample, producing an image from the transmitted electrons. Because of this, TEM can show details that are not clearly visible by SEM, especially the primary particles inside larger agglomerates. This is important for these samples because the powders can look strongly aggregated in SEM, but TEM allows to distinguish the smaller nanocrystalline particles and measure their size distribution more directly. In this way, TEM complements SEM/EMPA and XRPD, since it connects the external morphology and composition with the real particle size at the nanoscale²⁰.

3.2 Particle Size Distribution and Nanostructural Analysis

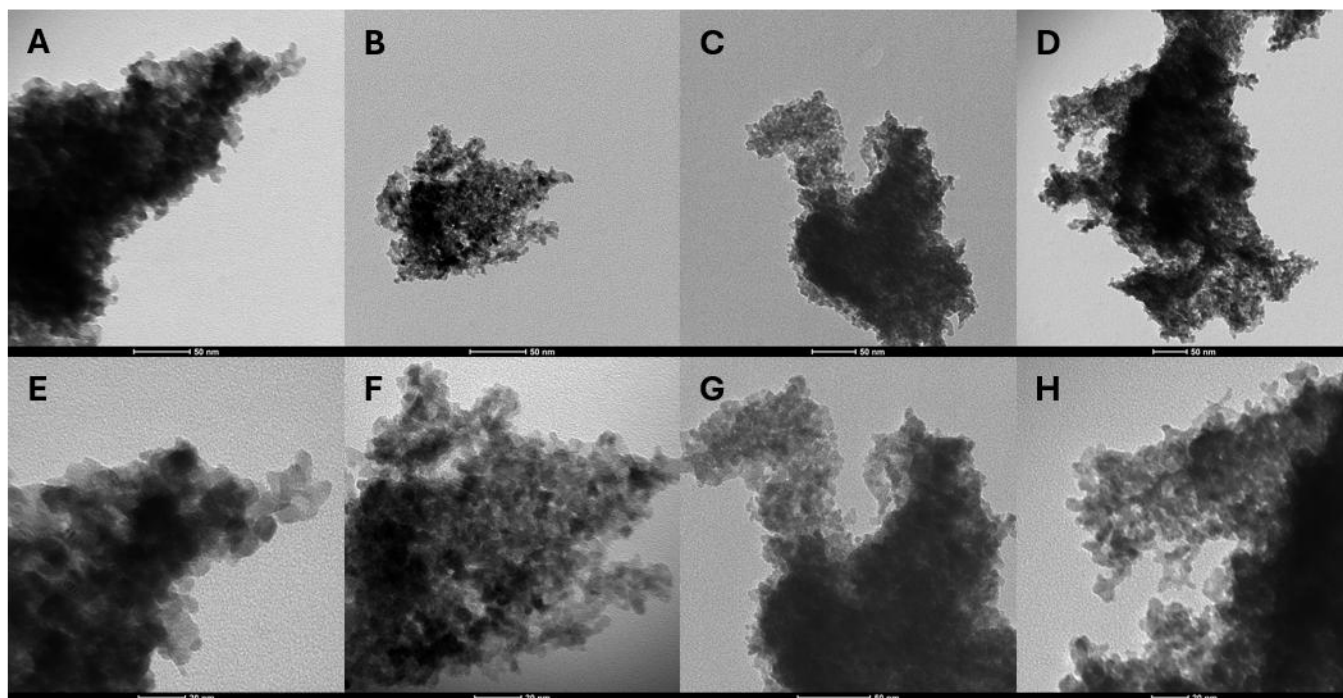


Figure 29. Bright-field TEM micrographs of the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ composites doped with increasing concentrations of Gd: A,E) $x_{\text{Gd}} = 0.130$; B,F) $x_{\text{Gd}} = 0.244$; C,G) $x_{\text{Gd}} = 0.383$; D,H) $x_{\text{Gd}} = 0.498$. Panels A-D show a magnification in the scale of 50 nm, while panels E-F display a higher magnification of the agglomerates at 20 nm, after heat treatment at 500 °C for 5 h.

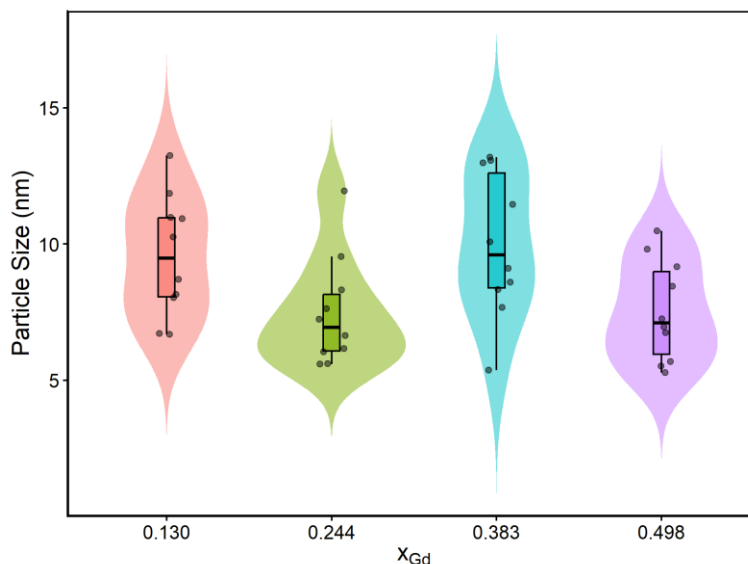


Figure 30. Particle size distribution represented by violin plots obtained from TEM analysis of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ composites doped with increasing concentrations of Gd.

The TEM micrographs show that the samples consist of nanocrystalline primary particles (Figure 29). These primary particles are not isolated, they form dense, irregular clusters or aggregates. The overall morphology remains relatively consistent across the different doping concentrations, showing a porous, sponge-like network typical of chemically synthesized nanoparticles.

The synthesized nanoparticles $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ had the following particle size distributions: $x_{\text{Gd}} = 0.130$: 9.6 ± 0.7 nm, $x_{\text{Gd}} = 0.244$: 7.5 ± 0.6 nm, $x_{\text{Gd}} = 0.383$: 10.0 ± 0.8 nm, and $x_{\text{Gd}} = 0.498$: 7.5 ± 0.6 nm (Figure 30). These distributions, represented by violin plots, were generated by measuring the maximum diameter of 10 randomly chosen nanoparticles per micrograph. In order to gauge any correlation between dopant concentration and its particle size distribution, only micrographs with a scale bar of 20 nm were used for statistical consistency during the analysis. This approach provides a perspective of the distribution of the nanoparticle size for the different the dopant concentration of Gd synthesized to observe if there exists any correlation between the nature of the sample in terms of dopant amount and particle size (Figure 30). This shows that, while the average sizes remained consistently under 10 nm, there is a significant spread in individual particle sizes. However, all samples remain within the same nanostructured regime, without a significant distinction on the average size for the different $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ samples.

4. X-Ray Powder Diffraction (XRPD)

4.1 Principle of the Technique

For understanding how the CeO_2 crystal structure changes when Gd is introduced into the lattice, XRPD analysis was performed on pure CeO_2 and in the different $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ samples.

Pure CeO_2 has a cubic fluorite structure. When Gd^{3+} ions are added, they can replace Ce^{4+} ions inside the structure. Since the charge of Gd^{3+} is lower than that of Ce^{4+} , oxygen vacancies are also formed to keep the material electrically neutral. For this reason, the diffraction pattern can give useful information about whether Gd is really incorporated into the CeO_2 lattice or if a secondary phase, such as Gd_2O_3 , is formed. The analysis was focused on three main points: the position of the diffraction peaks, the change in lattice parameter and unit-cell volume, and the broadening of the peaks. These aspects help to describe the structural effect of Gd doping on CeO_2 .

4.2 Structural Analysis of Gd-Doped CeO_2

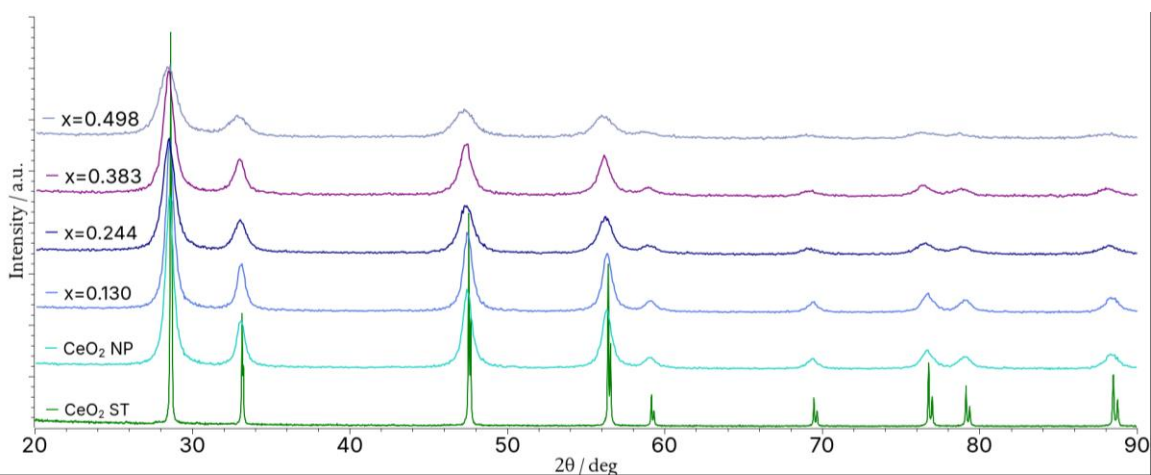


Figure 31. XRD patterns of pure CeO_2 (CeO_2 ST), nanoparticulate CeO_2 (CeO_2 NP) and composites $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ doped with increasing concentrations of Gd at $x_{\text{Gd}} = 0.130, 0.244, 0.383, 0.498$, respectively.

All patterns show the characteristic reflections of the cubic fluorite structure (Fm3m), with no extra peaks for secondary Gd_2O_3 phase, confirming that Gd^{3+} is successfully incorporated into the CeO_2 lattice rather than forming a separate phase (Figure 31). The relative peak intensities are preserved across all compositions, while peak heights decrease and peaks broaden somewhat with increasing Gd content, consistent with a reduction in crystallite size and an increase in lattice disorder upon doping.

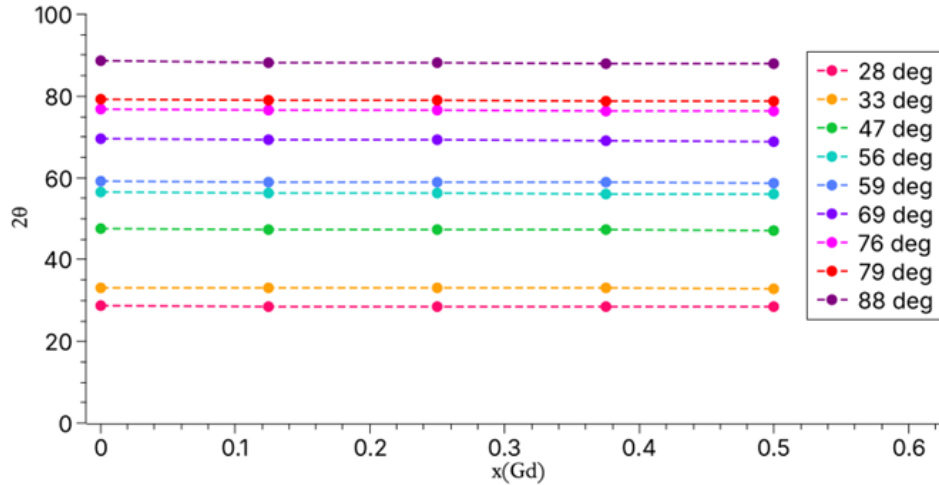


Figure 32. Peak shift (2θ position of the main fluorite reflections) as a function of Gd content, x_{Gd} .

The position of nine indexed reflections: 28° , 33° , 47° , 56° , 59° , 69° , 76° , 79° and 88° in 2θ are tracked across the doping series (Figure 32). All peaks remain essentially flat (constant 2θ) with increasing Gd content, with only a very slight tendency to shift to a lower angle. Taking into consideration that Gd^{3+} has a bigger ionic radius than Ce^{4+} , Bragg's law predicts that lattice expansion upon doping should shift reflections to lower 2θ ; which is confirmed as the small magnitude of the shift here indicates that the lattice effectively expanded.

To make the peak-shift analysis more quantitative, each reflection was fitted as a function of Gd content (Table 18). The slope gives an idea of how much the peak position changes with doping, while the intercept represents the extrapolated position at zero Gd content. The values of R^2 show that the points follow an almost linear behavior, although the total shift remains small.

Table 18. Reflection fitting parameters of Bragg peaks for $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ samples.

Bragg peak Reflection (2θ)	Parameter					
	m	δm	R^2	q	δq	δy
28	0.3459	0.0281	0.9806	28.6207	0.0086	0.0111
33	0.5528	0.0441	0.9812	33.1718	0.0135	0.0174
47	0.6782	0.0576	0.9788	47.5560	0.0176	0.0228
56	0.7102	0.0330	0.9936	56.4105	0.0101	0.0130
59	0.7533	0.0793	0.9678	59.1675	0.0243	0.0314
69	0.9887	0.0639	0.9876	69.4602	0.0196	0.0253
76	0.9366	0.0734	0.9819	76.7594	0.0225	0.0290
79	1.0098	0.0750	0.9837	79.1482	0.0230	0.0297
88	1.2792	0.1262	0.9716	88.4763	0.0386	0.0499

These results should be interpreted together. The shift in peak positions is indicative of cell expansion, since changes in the lattice spacing modify the diffraction angles. However, the refined lattice parameter and unit-cell volume provide a more direct and quantitative description of this effect. For this reason, the peak shift supports the structural effect of Gd doping, while the refined cell parameters give a more reliable measure of the lattice expansion.

The next step was to calculate the unit-cell volume for each composition (Figure 33). This parameter is more sensitive to small structural changes than looking only at the peak positions. If Gd is incorporated into the CeO_2 lattice, an increase in the cell volume is expected because Gd^{3+} has a larger ionic radius than Ce^{4+} .

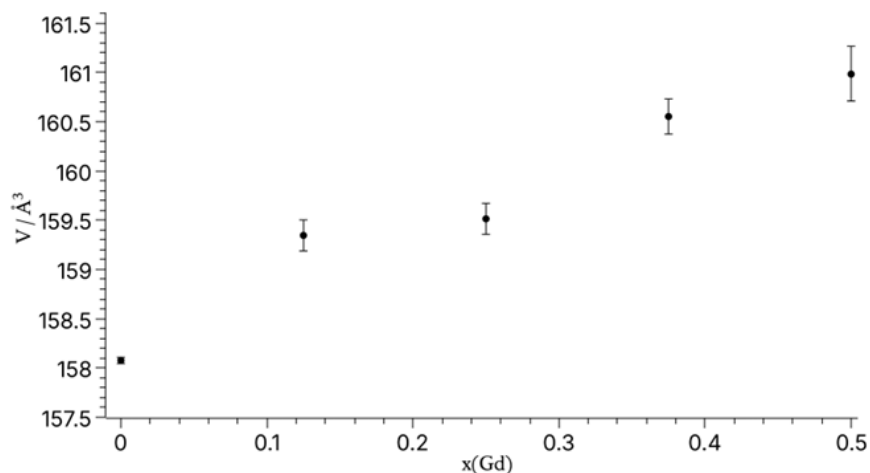


Figure 33. Unit-cell volume (V) of the fluorite lattice as a function of Gd content.

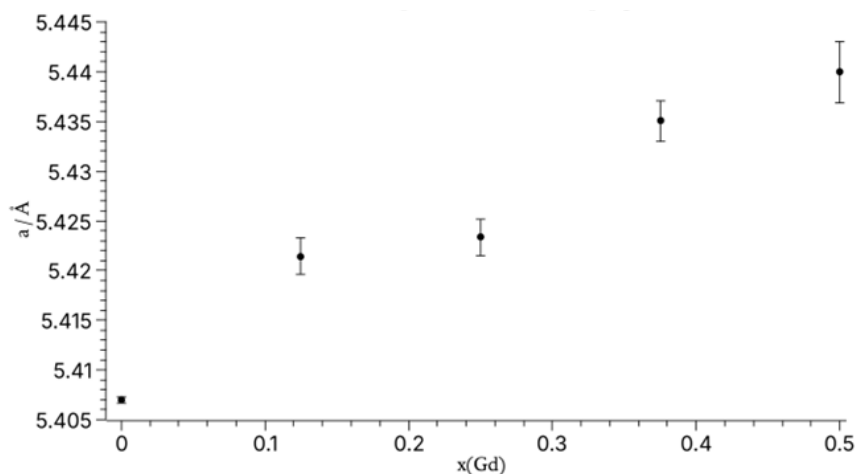


Figure 34. Lattice parameter (a) of the fluorite lattice as a function of Gd content.

Both the unit-cell volume and the cubic lattice parameter increase proportionally with increasing Gd content, from $a \approx 5.407 \text{ Å}$ ($V \approx 158.1 \text{ Å}^3$) for pure CeO_2 to $a \approx 5.440 \text{ Å}$ ($V \approx 161.0 \text{ Å}^3$) at $x = 0.498$ (Figure 34). This expansion is direct structural evidence that Gd^{3+} , with ionic radius $\approx 1.05 \text{ Å}$, is substituting for the smaller Ce^{4+} ($\approx 0.97 \text{ Å}$) on the cation sites of the fluorite lattice, in agreement with Vegard's-law-type behavior expected for a solid solution, and is consistent with the charge-compensating oxygen-vacancy mechanism described priorly²¹. In simple words, as more Gd enters the CeO_2 lattice, the unit cell expands gradually instead of showing the appearance of a new separated crystalline phase.

The following tables report the parameters obtained from peak fitting (Tables 19-24). These values describe the position and shape of the diffraction peaks for the standard CeO₂ sample, the nanoparticulate CeO₂ sample, and the Gd-doped compositions.

Table 19. Optimized parameters after peak fitting of CeO₂ Standard.

Index	2 θ / deg	Hg,G	Hg,L
1 1 1	28.6324 $\pm$ 0.0002	0.0368	0.04582
2 0 0	33.1671 $\pm$ 0.0004	0.02842	0.04768
2 2 0	47.5626 $\pm$ 0.0003	0.02801	0.04912
3 1 1	56.4165 $\pm$ 0.0003	0.0354	0.04941
2 2 2	59.1616 $\pm$ 0.0007	0.03791	0.04794
4 0 0	69.4783 $\pm$ 0.0007	0.03279	0.05383
3 3 1	76.7624 $\pm$ 0.0005	0.03351	0.05969
4 2 0	79.1352 $\pm$ 0.0006	0.04294	0.05324
4 2 2	88.4825 $\pm$ 0.0006	0.03704	0.07202

Table 20. Optimized parameters after peak fitting of CeO₂ nanoparticles.

Index	2 θ / deg	Hh,G	Hh,L	Hf,L	Hf,G	Hf	η	Bf / rad
1 1 1	28.634 $\pm$ 0.001	0.29529	0.44038	0.39456	0.29299	0.55612	0.76867	0.01374
2 0 0	33.169 $\pm$ 0.002	0.38109	0.36215	0.31447	0.38003	0.57383	0.62360	0.01334
2 2 0	47.589 $\pm$ 0.002	0.34962	0.48973	0.44061	0.34850	0.63918	0.75131	0.01567
3 1 1	56.456 $\pm$ 0.002	0.36394	0.52099	0.47158	0.36222	0.67489	0.75945	0.01660
2 2 2	59.214 $\pm$ 0.006	0.47759	0.39086	0.34292	0.47608	0.68319	0.57950	0.01561
4 0 0	69.534 $\pm$ 0.006	0.46976	0.53593	0.48209	0.46862	0.77597	0.69115	0.01855
3 3 1	76.840 $\pm$ 0.004	0.39220	0.66608	0.60639	0.39076	0.80538	0.80545	0.02021
4 2 0	79.216 $\pm$ 0.006	0.38619	0.67505	0.62180	0.38380	0.81225	0.81596	0.02048
4 2 2	88.586 $\pm$ 0.005	0.73179	0.52523	0.45321	0.73085	0.99905	0.53186	0.02240

Table 21. Optimized parameters after peak fitting of Ce_{1-x}Gd_xO_{2-(x/2)}, for which x_{Gd}=0.130.

Index	2 θ / deg	Hh,G	Hh,L	Hf,L	Hf,G	Hf	η	Bf / rad
1 1 1	28.565 $\pm$ 0.002	0.2197	0.5093	0.4635	0.2166	0.5527	0.8754	0.0143
2 0 0	33.093 $\pm$ 0.004	0.2289	0.4997	0.4520	0.2272	0.5506	0.8614	0.0142
2 2 0	47.444 $\pm$ 0.003	0.1815	0.5910	0.5419	0.1793	0.5983	0.9284	0.0159
3 1 1	56.308 $\pm$ 0.001	0.2021	0.6073	0.5579	0.1990	0.6245	0.9188	0.0165
2 2 2	59.05 $\pm$ 0.01	0.2042	0.6121	0.5642	0.2007	0.6311	0.9192	0.0167
4 0 0	69.30 $\pm$ 0.01	0.1757	0.6523	0.5985	0.1726	0.6467	0.9436	0.0173
3 3 1	76.63 $\pm$ 0.01	0.4163	0.6227	0.5630	0.4149	0.7909	0.7707	0.0196
4 2 0	79.01 $\pm$ 0.01	0.2921	0.7390	0.6857	0.2890	0.7959	0.8938	0.0208
4 2 2	88.269 $\pm$ 0.009	0.2759	0.8215	0.7495	0.2734	0.8426	0.9158	0.0222

Table 22. Optimized parameters after peak fitting of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$, for which $x_{\text{Gd}}=0.244$.

Index	2θ / deg	Hh,G	Hh,L	Hf,L	Hf,G	Hf	η	Bf / rad
1 1 1	28.532 ± 0.003	0.48343	0.67679	0.63098	0.48202	0.90077	0.76094	0.02217
2 0 0	33.051 ± 0.006	0.45489	0.70045	0.65277	0.45400	0.89434	0.78606	0.02225
2 2 0	47.413 ± 0.005	0.66123	0.67823	0.62911	0.66064	1.05681	0.66753	0.02502
3 1 1	56.246 ± 0.006	0.49361	0.86073	0.81132	0.49234	1.05309	0.82001	0.02659
2 2 2	59.00 ± 0.02	0.55243	0.83909	0.79115	0.55113	1.08466	0.78564	0.02699
4 0 0	69.22 ± 0.02	0.83616	0.71799	0.66416	0.83552	1.24239	0.61086	0.02874
3 3 1	76.56 ± 0.01	0.98983	0.66186	0.60217	0.98927	1.34477	0.52599	0.03008
4 2 0	78.94 ± 0.02	0.99356	0.67340	0.62016	0.99263	1.35999	0.53423	0.03052
4 2 2	88.23 ± 0.02	0.56264	1.19100	1.11898	0.56142	1.36225	0.86172	0.03504

Table 23. Optimized parameters after peak fitting of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$, for which $x_{\text{Gd}}=0.383$.

Index	2θ / deg	Hh,G	Hh,L	Hf,L	Hf,G	Hf	η	Bf / rad
1 1 1	28.487 ± 0.002	0.35572	0.64077	0.59495	0.35381	0.76652	0.82477	0.01940
2 0 0	32.979 ± 0.005	0.34235	0.62676	0.57908	0.34116	0.74355	0.82695	0.01883
2 2 0	47.305 ± 0.004	0.33775	0.74251	0.69340	0.33658	0.83589	0.86825	0.02156
3 1 1	56.134 ± 0.005	0.21284	0.82497	0.77556	0.20988	0.83104	0.94958	0.02225
2 2 2	58.92 ± 0.02	0.21053	0.82006	0.77212	0.20709	0.82644	0.95037	0.02213
4 0 0	69.10 ± 0.02	0.10289	0.91696	0.86313	0.09753	0.87487	0.98997	0.02387
3 3 1	76.37 ± 0.01	0.30364	0.94763	0.88795	0.30178	0.98491	0.92515	0.02607
4 2 0	78.76 ± 0.02	0.31919	0.97789	0.92465	0.31629	1.02681	0.92433	0.02717
4 2 2	87.98 ± 0.02	0.47983	0.99705	0.92503	0.47840	1.13672	0.85553	0.02916

Table 24. Optimized parameters after peak fitting of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$, for which $x_{\text{Gd}}=0.498$.

Index	2θ / deg	Hh,G	Hh,L	Hf,L	Hf,G	Hf	η	Bf / rad
1 1 1	28.455 ± 0.004	0.59064	0.97980	0.93398	0.58949	1.23047	0.81056	0.03095
2 0 0	32.88 ± 0.01	0.59885	1.10116	1.05348	0.59817	1.33514	0.83537	0.03395
2 2 0	47.208 ± 0.009	0.92442	0.92570	0.87658	0.92399	1.47568	0.66636	0.03492
3 1 1	56.06 ± 0.01	0.60285	1.21103	1.16162	0.60181	1.42824	0.85517	0.03663
2 2 2	58.76 ± 0.04	0.61290	1.23121	1.18327	0.61173	1.45389	0.85561	0.03730
4 0 0	68.96 ± 0.06	1.07668	1.03387	0.98004	1.07618	1.68906	0.65367	0.03976
3 3 1	76.30 ± 0.04	1.01213	1.22976	1.17007	1.01157	1.77201	0.72595	0.04298
4 2 0	78.63 ± 0.05	1.01209	1.22972	1.17648	1.01118	1.77653	0.72764	0.04312
4 2 2	87.83 ± 0.05	1.24151	1.19214	1.12012	1.24095	1.94047	0.65091	0.04562

The most important information is not every individual number, but the general evolution of the peak width and peak shape. Broader peaks usually indicate smaller crystallite size, higher disorder, or both. For this reason, these fitted parameters were later used in the Williamson-Hall analysis.

After fitting the diffraction peaks, the peak broadening was analyzed using the Williamson-Hall method (Figure 35). This procedure is useful because the width of a diffraction peak is not only

related to small crystallite size, but also to lattice strain. In a simplified form, the total broadening can be written as the sum of a size contribution and a strain contribution:

$$\beta_{(hkl)} \cos \theta_{(hkl)} = \frac{\lambda}{D_v} + 4\varepsilon \sin \theta_{(hkl)}$$

where β is the corrected peak width in radians, θ is the Bragg angle, λ is the X-ray wavelength, D is the apparent crystallite size and ε is the microstrain. In this way, a plot of $\beta \cos \theta$ against $4 \sin \theta$ gives a straight line, where the intercept is $\frac{\lambda}{D}$ and the slope corresponds to ε . Therefore, smaller intercept values indicate larger crystallites, while a larger slope indicates more lattice deformation or microstrain inside the material²².

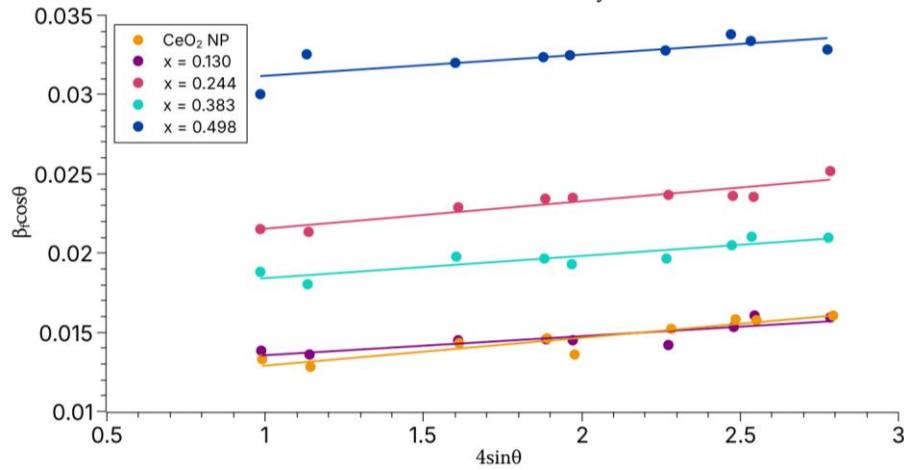


Figure 35. Williamson-Hall plot ($\beta \cos \theta$ vs. $4 \sin \theta$) for CeO_2 and the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ compositions.

For the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ samples, this analysis shows that Gd doping produces broader diffraction peaks and smaller crystallite sizes. The crystallite size decreases from around 12.5 nm for pure CeO_2 to about 4.7 nm for the most doped sample, while the microstrain remains in a relatively similar range (Figure 36). This suggests that Gd incorporation and the related oxygen vacancies mainly limit crystal growth during synthesis, rather than producing a very strong increase of strain. In this sense, the Williamson-Hall results support the XRPD interpretation: the samples remain as fluorite-type solid solutions, but the lattice becomes more defective and nanostructured as the Gd content increases.

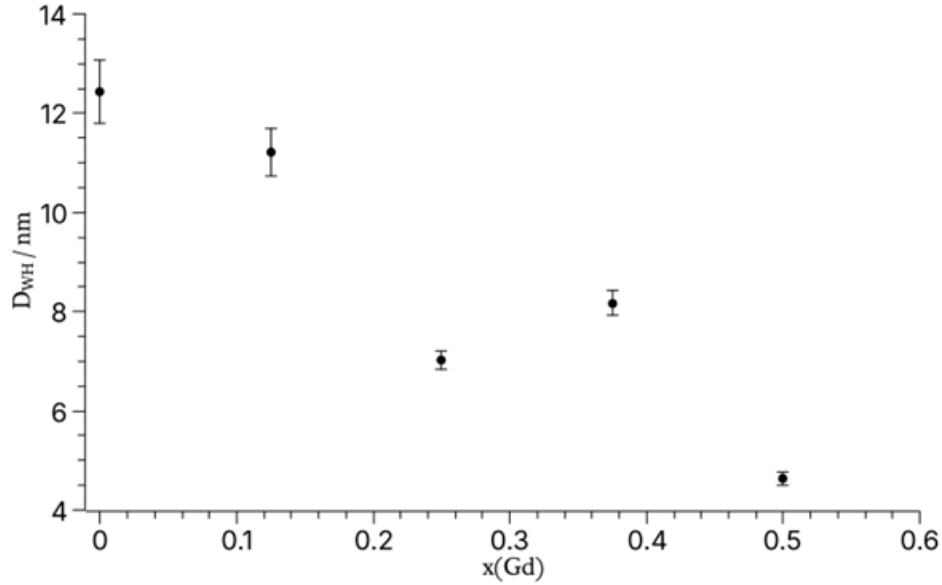


Figure 36. Crystallite size (D_{WH}) extracted from the Williamson-Hall analysis as a function of Gd content.

The crystallite size decreases from about 12.5 nm in undoped CeO_2 to a minimum of about 4.7 nm at $x_{\text{Gd}} = 0.498$, with an essentially monotonic downward trend (apart from a small local increase at $x_{\text{Gd}} = 0.383$). This refinement of the crystallite size with increasing dopant concentration is commonly observed in doped sol-gel oxides, since the incorporation of a foreign cation and the associated lattice strain inhibit crystallite growth during the calcination process.

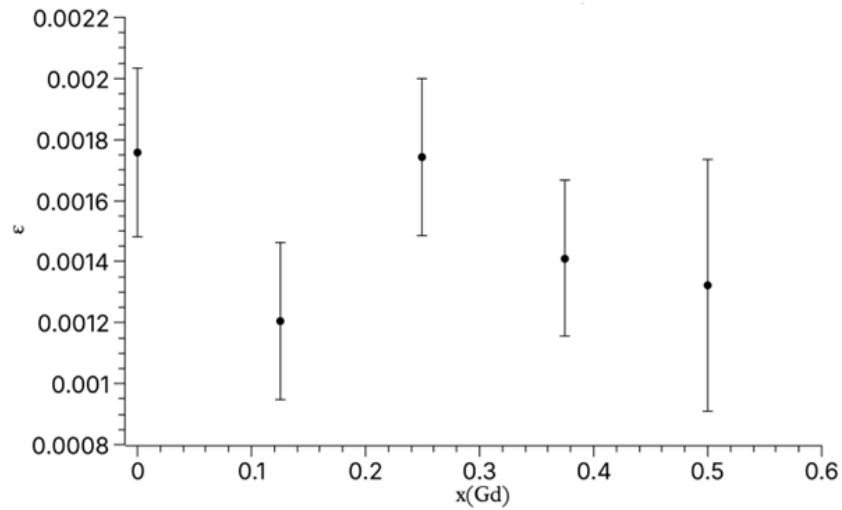


Figure 37. Microstrain (ϵ) as a function of Gd content.

The microstrain shows more scatter than the crystallite size, fluctuating between about 0.0012 and 0.0018 with overlapping error bars across the doping series, and no strong systematic trend with composition (Figure 37). This suggests that, within experimental uncertainty, the lattice strain is not strongly affected by Gd substitution and that the dominant structural effect of doping is the reduction in crystallite size (Figure 36) together with the lattice expansion (Figures 33-34), rather than a build-up of microstrain.

As a summary, the XRPD results show that Gd^{3+} is successfully and homogeneously incorporated into the CeO_2 fluorite lattice without forming a secondary phase. Doping produces a clear lattice expansion, with larger a parameter and Volume, consistent with the larger ionic radius of Gd^{3+} , while simultaneously refining the crystallite size, consistent with dopant-induced inhibition of grain growth. Microstrain remains comparatively constant across the series, indicating that the lattice accommodates the dopant mainly through volume expansion and oxygen-vacancy formation rather than through a large build-up of internal strain.

5. UV-Visible Diffuse Reflectance Spectroscopy (UV/Vis-DRS)

5.1 Principle of the Technique

UV-Vis diffuse reflectance spectroscopy was used here under the same general idea described for AgI: the powders were analyzed through their reflected light as a function of wavelength, and the reflectance data were treated with the Kubelka-Munk function, $F(R_\infty) = (1 - R_\infty)^2 / 2R_\infty$, to obtain a quantity related to optical absorption. The band gap was then estimated from Tauc plots by extrapolating the linear region of $(F(R_\infty) \cdot E)^2$ versus photon energy, using the same direct-transition treatment applied in the previous UV-Vis section.

However, the interpretation is different in the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ system, as CeO_2 keeps the cubic fluorite-type structure, and the main variable is not a phase transition but the progressive substitution of Ce^{4+} by Gd^{3+} . This substitution introduces oxygen vacancies for charge compensation and modifies the local disorder of the oxide lattice. Therefore, changes in the absorption edge or in the apparent optical band gap should be read mainly as the effect of Gd incorporation, oxygen vacancy-related states, crystallite size, and structural disorder, rather than as evidence of a new crystal phase.

5.2 Optical Response and Band-Gap Evolution in Gd-Doped CeO_2

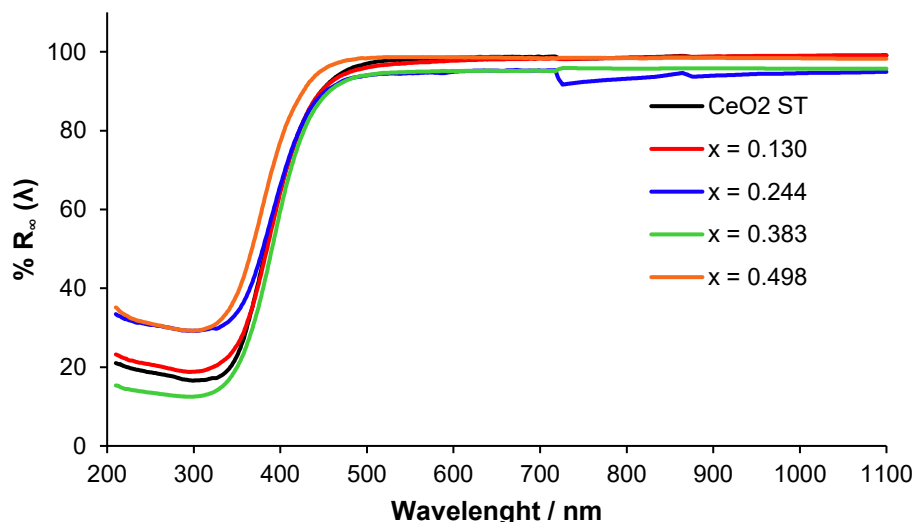


Figure 38. UV/Vis-DRS spectrum of CeO_2 and the $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ compositions.

All compositions show high reflectance ($\approx 90\text{-}100\%$) above ~ 450 nm and a sharp drop in reflectance below ~ 400 nm, marking the fundamental absorption edge of the fluorite oxide (Figure 38). The position and steepness of this edge vary slightly between compositions, particularly for $x_{\text{Gd}} = 0.383$, which shows a more gradual decrease and lower reflectance minimum than the other samples, foreshadowing the band-gap differences quantified below.

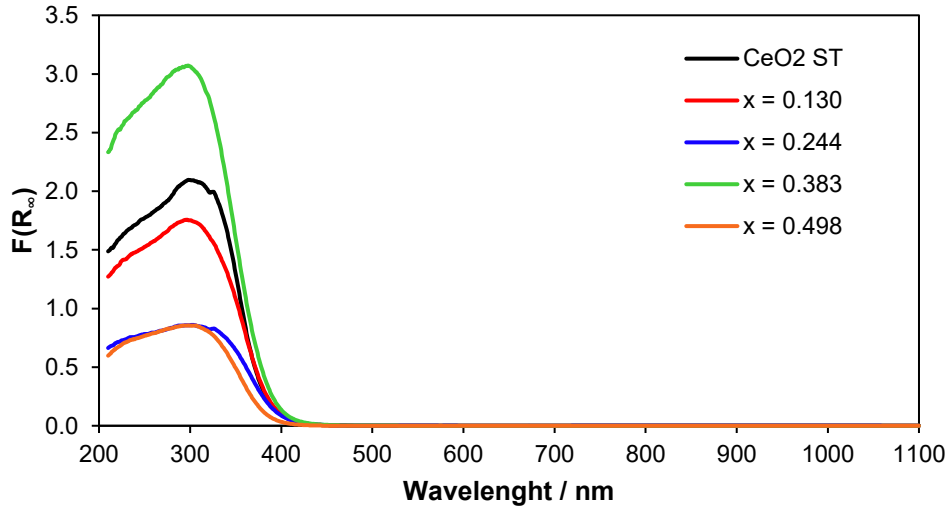


Figure 39. Kubelka-Munk plot, $F(R_{\infty})$, of CeO_2 and the Gd-doped CeO_2 compositions.

The Kubelka-Munk transformation converts the reflectance spectra into absorption-proportional curves (Figure 39). All compositions show a clear absorption maximum near 280-320 nm. Notably, the $x_{\text{Gd}} = 0.383$ sample shows the highest $F(R_{\infty})$ intensity (peak ≈ 3.1), markedly larger than the other compositions, indicating a stronger optical absorption in this sample, which is consistent with a higher concentration of oxygen-vacancy-related absorbing centres (F-centres) at this doping level.

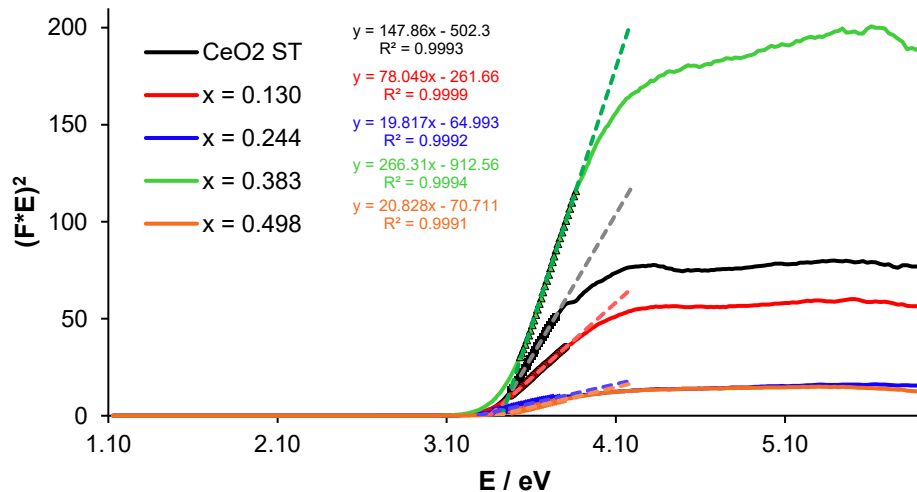


Figure 40. Tauc plot as a function of photon energy for CeO_2 and $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-(x/2)}$ compositions (full energy range).

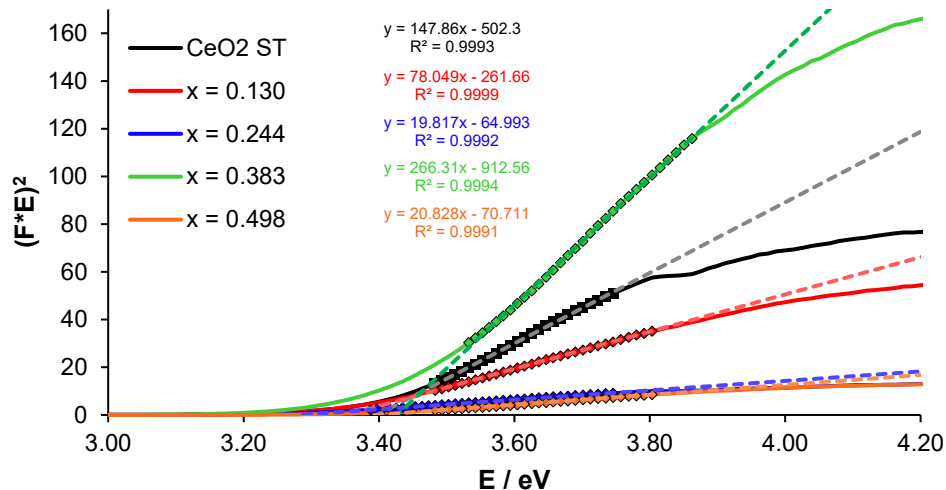


Figure 41. Magnified view of the Tauc plot in the region of the absorption edge, with linear extrapolations used to determine the band-gap energy.

The Tauc plots (Figures 40-41) show the characteristic steep rise of $(F(R_{\infty}) \cdot E)^2$ near the absorption edge for all compositions. Linear fits to this rising region were extrapolated to the energy axis to obtain the optical band gap for each composition; the corresponding band-gap values are summarized below (Table 25).

Table 25. Relationship between dopant concentration and optical bandgap energy (E_g) for $\text{Ce}_{x-1}\text{Gd}_x\text{O}_{2-(x/2)}$.

x_{Gd}	E_g (eV)
0.000	3.40
0.125	3.35
0.250	3.28
0.375	3.43
0.500	3.39

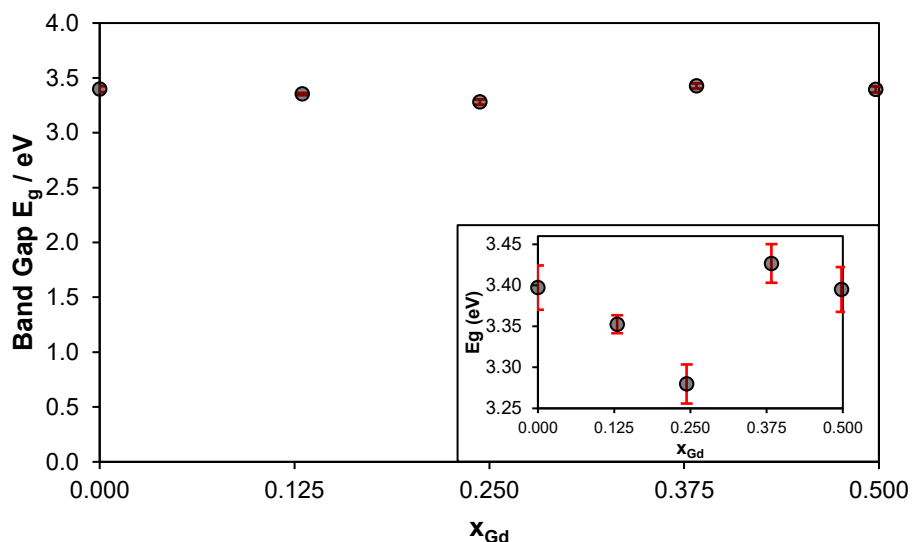


Figure 42. Band-gap energy (E_g) as a function of Gd content, x_{Gd} , for $Ce_{1-x}Gd_xO_{2-(x/2)}$ (inset: magnified view).

From figure 42 we observed that the band gap does not change with Gd content. Instead, The band gaps obtained display an non-correlating behavior as the values align around an horizontal line, with only some local tendencies portrayed making a significant magnification that can be attributed to random variations in the measurements.

In summary, the optical band gap did not change in a simple linear way with Gd content. This behavior can be understood as the result of two effects acting at the same time: the creation of defect-related states, which can reduce the apparent gap, and changes in crystallite size or structural disorder, which can shift the absorption edge in the opposite direction.

6. Raman Spectroscopy

6.1 Principle of the Technique

Raman spectroscopy was used to study the local structure and vibrational behavior of the $Ce_{1-x}Gd_xO_{2-(x/2)}$ system. This technique is based on the interaction between a monochromatic laser and the sample, where most of the light is scattered without changing energy, but a very small fraction interacts with lattice vibrations and is scattered with a different energy. This energy difference is reported as the Raman shift, usually in cm^{-1} , and gives information about the vibrational modes of the crystal²³. For this reason, Raman is especially useful to detect changes in local symmetry, disorder, and defects that may not be clearly visible by XRPD. While XRPD describes the average long-range crystal structure, Raman is more sensitive to what happens locally around the ions.

In pure CeO_2 , the fluorite structure normally gives one strong Raman band close to 465 cm^{-1} , assigned to the F_{2g} vibrational mode, mainly related to the symmetric vibration of oxygen atoms around Ce^{4+} . When Ce^{4+} is partially replaced by Gd^{3+} , oxygen vacancies are formed to maintain charge balance, and these vacancies disturb the local arrangement of the oxygen sublattice. Therefore, changes such as peak broadening, shifts in the F_{2g} position, loss of intensity, or the

appearance of additional bands can be associated with increasing structural disorder and defect formation in the doped samples. In this sense, Raman spectroscopy complements XRPD by confirming whether the fluorite structure is locally affected by Gd incorporation, even when the average crystalline phase remains the same.

6.2 Gd-Doped CeO₂ (powder)

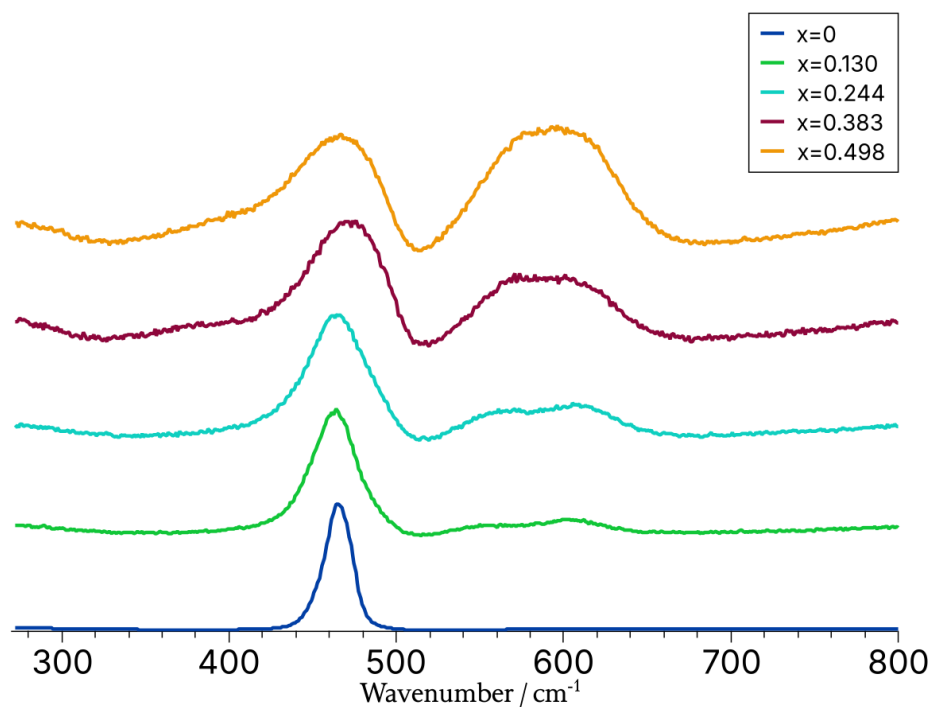


Figure 43. Raman spectra of Ce_{1-x}Gd_xO_{2-(x/2)} regular powders with different Gd contents.

The Raman spectra of the regular powder samples show a clear evolution with Gd content (Figure 43). For pure CeO₂, the main signal is a sharp peak close to 465 cm⁻¹, which is typical of the fluorite phase. As the amount of Gd increases, this peak becomes broader and less intense. This means that the local order of the CeO₂ lattice is progressively affected by the dopant.

The broadening of the main peak is important because it indicates that the oxygen environment around the cations is no longer as regular as in pure CeO₂. This is expected, because the substitution of Ce⁴⁺ by Gd³⁺ produces oxygen vacancies. These vacancies break part of the local symmetry and generate disorder in the lattice. Also at higher Gd content, additional broad bands appear, especially in the region around 550-600 cm⁻¹. These bands can be associated with defect-related vibrations, mainly connected with oxygen vacancies and local cation-vacancy arrangements.

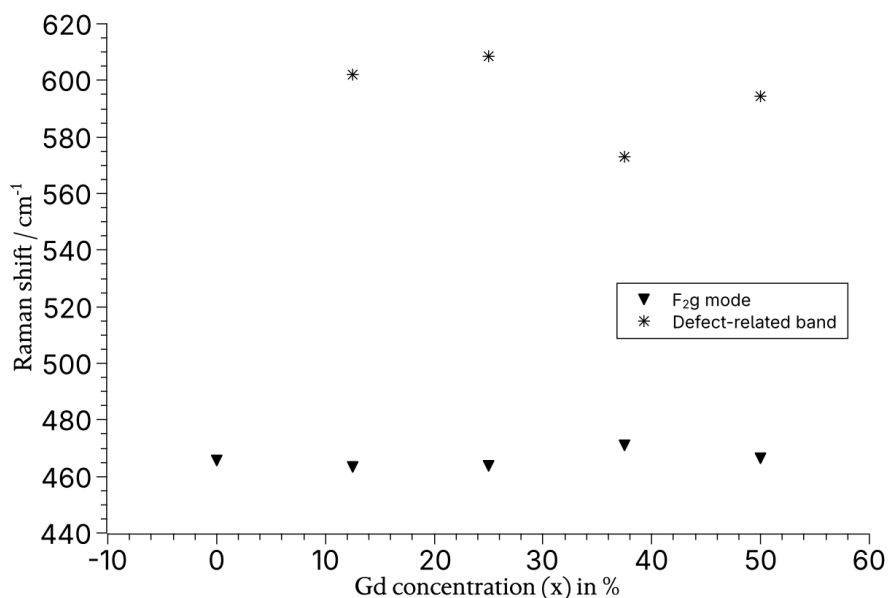


Figure 44. Raman peak positions of regular $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-x/2}$ powders as a function of Gd content.

To follow these changes more clearly, the position of the main Raman bands was plotted as a function of Gd concentration (Figure 44). The F_{2g} mode remains close to 465 cm^{-1} , but it shows a small displacement when Gd is introduced. This shift is not very large, but it is consistent with the incorporation of Gd into the CeO_2 lattice. The small redshift of the F_{2g} band can be explained by the formation of oxygen vacancies and by the change in the local coordination around the cations. At the same time, the larger ionic radius of Gd^{3+} compared with Ce^{4+} can contribute to lattice expansion. These two effects act together, so the final shift is moderate rather than very large. The high-frequency band becomes more visible as Gd content increases.

This supports the idea that the doped samples contain more defect-related environments than pure CeO_2 .

6.3 Gd-Doped CeO_2 (pellets)

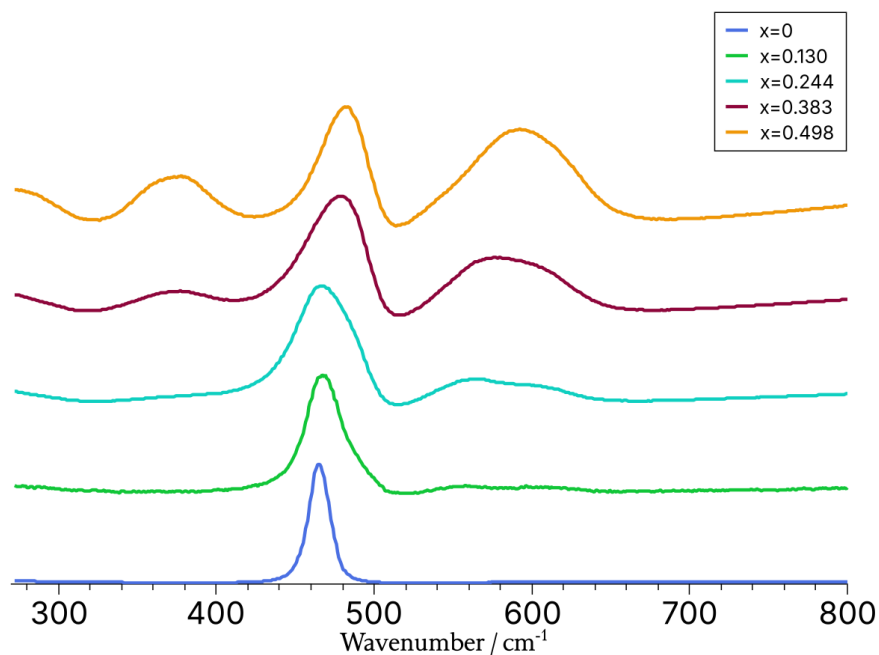


Figure 45. Raman spectra of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-x/2}$ pellets with different Gd contents.

The same Raman analysis was also performed on pellet samples (Figure 45). This comparison is useful because the pellet preparation can affect the microstructure of the material. Pressing and thermal treatment can modify grain contact, local stress, and defect distribution. Therefore, even if the chemical composition is the same, the Raman spectrum can show small differences between powders and pellets. In general, the pellet spectra show the same main features as the regular powders: the F_{2g} band near 465 cm^{-1} and the defect-related bands at higher wavenumbers. This means that the main fluorite-type structure is still present after pellet preparation.

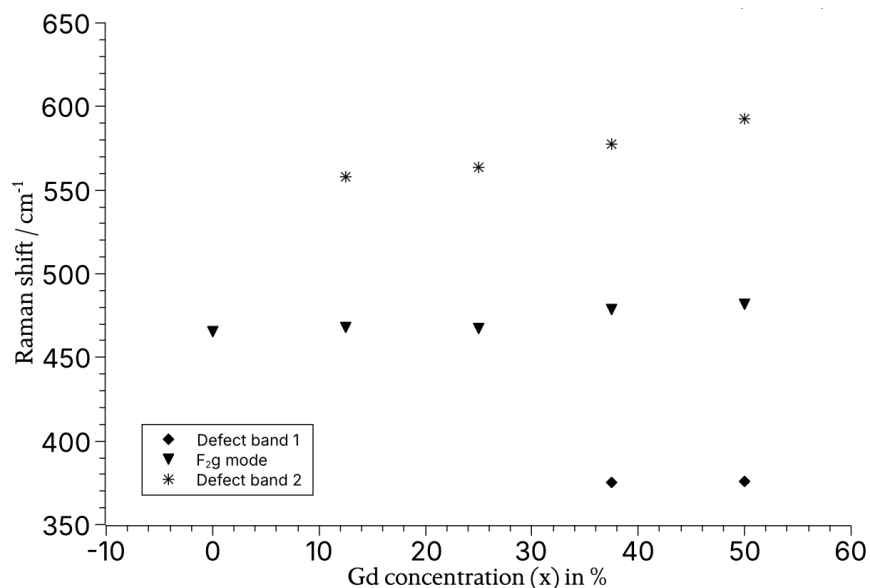


Figure 46. Raman peak positions of $\text{Ce}_{1-x}\text{Gd}_x\text{O}_{2-x/2}$ pellets as a function of Gd content.

The peak-position plot for the pellets confirms that the main F_{2g} mode remains close to the same region observed for the powder samples (Figure 46). This indicates that pellet formation does not completely change the crystal structure. However, some bands appear broader or slightly displaced, especially at higher Gd content.

The differences between powders and pellets can be mainly attributed to the thermal treatment used to prepare the pellets. Since the pellets were fired at 900 °C, their microstructure is expected to differ from that of the original powders, due to particle growth, sintering, and possible changes in the defect structure. These factors can affect the Raman modes, especially their position, width, and relative intensity. Therefore, the comparison should be interpreted considering the different microstructural state of the samples.

Comparing regular powders and pellets, both sets of samples show the same general Raman behavior. Pure CeO_2 has a sharp F_{2g} band, while Gd-doped samples show broader peaks and stronger defect-related contributions. In other words, the main structural effect comes from Gd doping, not only from the physical form of the sample. The most important result is that Raman confirms the presence of local disorder in the doped samples. This agrees with the available results for XRPD analysis, where the fluorite structure was preserved, but the lattice parameter increased with Gd content. Together, XRPD and Raman show that Gd enters the CeO_2 lattice, creates oxygen vacancies, and modifies the local structure without producing a clearly separated crystalline secondary phase.

7. Overall Conclusion

The Gd-doped CeO_2 samples show that the Pechini synthesis was effective to obtain a doped ceria system where Gd enters mostly into the fluorite CeO_2 structure instead of forming a separated crystalline Gd_2O_3 phase. SEM and TEM show that the samples are nanostructured, with small primary particles grouped into bigger agglomerates, and with a morphology that becomes more porous and granular after doping. XRPD confirms that all compositions keep the cubic fluorite structure, but the lattice parameter and cell volume increase with Gd content, which is consistent with the replacement of Ce^{4+} by the larger Gd^{3+} ion and the formation of oxygen vacancies for charge compensation.

The optical and Raman results also support this interpretation. The UV-Vis band gap does not change in a perfectly linear way: it decreases at intermediate Gd concentration and then increases again for higher doping, which can be explained by the competition between defect states, oxygen vacancies, disorder and crystallite size effects. Raman spectroscopy shows that the main fluorite F_{2g} band remains present, so the basic CeO_2 structure is preserved, but the broadening of the band and the appearance of defect-related features indicate that the local structure becomes more disordered with Gd incorporation. Therefore, the most important conclusion is that Gd doping modifies CeO_2 from inside the lattice: it expands the cell, introduces vacancies and disorder, changes the morphology, and finally affects the optical response of the material.

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